Can Satellites Replace the Field Plot? A Blind Benchmark of Satellite Forest Carbon Against Standard-Formula Ground Truth, and Conformal Crediting with Calibrated Over-Crediting Risk

Kenny Tang
Computer Science Department
Bina Nusantara University
Jakarta, Indonesia 11480
kenny.tang@binus.ac.id

Riki Awal Syahputra
Computer Science Department
Bina Nusantara University
Jakarta, Indonesia 11480
riki.syahputra@binus.ac.id

Richard Hans
Computer Science Department
Bina Nusantara University
Jakarta, Indonesia 11480
richard.hans001@binus.ac.id

Evelyn Jane Sutjiadi
Computer Science Department
Bina Nusantara University
Jakarta, Indonesia 11480
evelyn.sutjiadi@binus.ac.id

Yenny Chen
Computer Science Department
Bina Nusantara University
Jakarta, Indonesia 11480
yenny.chen@binus.edu

Jude Joseph Lamug Martinez
Computer Science Department
Bina Nusantara University
Jakarta, Indonesia 11480
jmartinez@binus.edu

***Abstract*—Satellite platforms promise to replace field inventories in forest carbon crediting, but their estimates are rarely tested blind against the standard-formula answer, and credits are issued with deductions whose risk is unknown. We separate the question into two isolated grids. The right grid computes field ground truth to SNI 7724, IPCC, CDM, and Verra rules with a Monte Carlo uncertainty budget and is sealed. The left grid sees only plot coordinates and dates and estimates carbon from time-matched Sentinel-1/2, Landsat, PALSAR, and terrain data. A hash-chained ledger makes changes after the reveal detectable. On 476 stock plots, blind error was 29% on 1 ha plots with Sentinel, Landsat, and PALSAR, and 21% after adding a 1 m canopy height map and GEDI lidar. Project totals came within 7–9% outside Indonesia, while small Indonesian plots stayed near 100%. Counting trees on the 1 m canopy height map and applying the field formula found 3% of the stems and 25% of the carbon, and in 1,000 simulated projects the best map found a median 95% of the field carbon. Only 21.5% of Sentinel-2 acquisitions see an Indonesian plot clear, and a common scene-level filter discards 65% of those. We then propose conformal crediting: a project is credited a lower bound calibrated on other projects with field truth. Under cross-region transfer it met a 10% over-crediting target for each of five maps, within 2.9 points. It was invariant to map inflation, and credited 72–84% of the true stock without a field plot. The VM0047 deduction over-credited 23–29% of projects with field plots and sits below the integrity-yield frontier. For annual growth on 196 Bornean plots, no satellite estimate beat assuming no change. Satellites can map stock and support calibrated crediting; growth still needs remeasurement.**

***Index Terms*—forest carbon, MRV, SNI 7724, VM0047, remote sensing, cloud cover, Sentinel, ALOS PALSAR, machine learning, spatial cross-validation, conformal prediction, uncertainty, model-assisted estimation, commit-reveal ledger, distributed processing**

I. Introduction

A. Two Ways to Measure Forest Carbon

Traditional MRV (measurement, reporting, and verification) is field based. Crews lay out plots, measure each stem's diameter D and height H, identify species to assign wood density ρ, and pass these inputs through a published allometric equation [1]. Carbon follows from a fixed carbon fraction and the CO_2 to C mass ratio. Methodologies such as Verra VM0047 prescribe this chain [2]; its output is deterministic and is the number on which credits are issued. Nobody can change the formula: it is the international standard. What is costly is the visit, especially in remote Indonesian peat swamps and hill forests.

System-based MRV replaces or supplements the visit with satellites and software. Optical sensors observe canopy greenness, radar observes canopy structure and moisture, and spaceborne lidar measures height [3]. Global biomass maps exist [4], [5], and commercial platforms, including our own NusaCarbon prototype, turn polygons into carbon numbers. The open question is how close such numbers are to what the field formula would have produced at the same place and time.

B. Why the Comparison Is Hard to Do Honestly

Three traps inflate reported accuracy. First, *leakage*: if a model is trained or tuned on the plots it is scored on, or on plots next to them, the score measures memory rather than skill; spatially blocked validation of large-scale biomass maps reveals much lower skill than random validation [6], [7]. Second, *temporal mismatch*: a plot measured in 2009 and an image from 2023 describe different forests. Third, *scale*: a

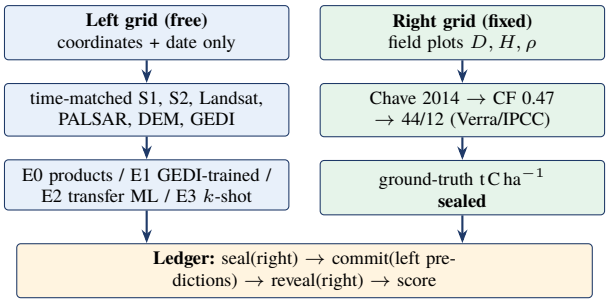


Fig. 1. Blind two-grid design. Only the public manifest (plot id, coordinates, date, radius) crosses from right to left.

plot is a fraction of a hectare while satellite pixels range from 10 m to 4.6 km. A fair test must isolate the estimator from the answer, match time explicitly, and report how error changes with scale.

Our own platform illustrates the need. An audit of the NusaCarbon code base found that its forest and mangrove figures depended mostly on default constants (for example, a fixed sequestration rate and tree density bins from NDVI) rather than on satellite measurement, that clouds were screened only at scene level, that the reported confidence was a grade computed from image metadata, and that no estimate had been compared with field data. This paper builds the comparison that was missing, and tests the cloud handling and the confidence statement as carefully as the estimate itself.

C. Two Isolated Grids

We separate the problem into two grids (Fig. 1). The **right grid** is fixed: published field plots, the standard formula, a carbon density in t C ha^{-1}. We never modify it. The **left grid** is free: any satellite data, any algorithm, any machine-learning method, subject to one rule: it never sees the right grid's outcomes for the plots it is scored on. A ledger enforces the rule and makes it auditable.

D. Research Questions

RQ1 (blind accuracy; research methodology). How far does a blind, satellite-only estimate of aboveground carbon stock deviate from standard-formula field ground truth, and how does the deviation depend on the temporal gap between field and satellite observations and on the area over which it is aggregated?

RQ2 (closing the gap; AI). Which learning strategy brings the satellite estimate closest to the field answer: off-the-shelf global products, a model trained only on spaceborne lidar, a model transferred from other regions, or a model calibrated with k local field plots? Are its prediction intervals honest? And can trees be counted from space, so that the field formula itself is applied to satellite-derived trees?

RQ3 (clouds and confidence; AI and methodology). How often do clouds hide an Indonesian plot from optical satellites, how much accuracy does that cost, and can the estimate and its prediction interval stay honest when they do? At the scale of a project, does the stated uncertainty protect against over-crediting, and how many field plots does a satellite save?

RQ4 (sequestration). Can the annual change in carbon stock, the quantity credits are issued on, be estimated blind from satellites, and how certain is the field answer itself?

RQ5 (integrity; AI and methodology). Can a credit be issued from a satellite estimate with a stated, verified risk of over-crediting, whatever map it comes from, and how many credits does that leave compared with the registry deduction and with field plots?

RQ6 (distributed systems). Can time-matched extraction for hundreds of plots and sensors be sharded, balanced, and made fault tolerant with bit-identical outputs, and can a hash-chained ledger make the order of prediction and reveal verifiable?

E. Contributions

(i) A blind, time-matched evaluation protocol with a commit-reveal ledger, reusable for any MRV estimator. (ii) A right grid built step by step to SNI 7724, IPCC, CDM, Verra, and Gold Standard rules, with a Monte Carlo uncertainty budget, on 476 stock plots (148 Indonesian plots recomputed tree by tree) and 223 day-dated repeat-census plots in Borneo. (iii) An estimator ladder from zero field data to k local plots, with spatially blocked validation. (iv) A plot-level Sentinel-2 clear-sky climatology for Indonesian forest plots and a measured error rate of the scene-level cloud filter. (v) A test of missing-data-augmented conformal intervals [8] under cloud-driven, non-random loss of the optical block with out-of-region calibration. (vi) An empirical coverage test of project-level intervals under spatial transfer, and a simulation of how often the VM0047 uncertainty deduction still over-credits. (vii) *Conformal crediting*: a crediting rule calibrated across projects against field truth. It meets its over-crediting target under spatial transfer and for biased maps, it is invariant to uniform map inflation, and it comes with an integrity-yield frontier that prices a field plot in credits. (viii) A tree-by-tree satellite estimator (E4) that counts trees on a 1 m canopy height map and applies the field formula, benchmarked against field stem lists and a wood-density classification of the field stems, and 1,000 project simulations reported as carbon amounts, with every plot row (Appendix A). (ix) A sharded, idempotent extraction runner with measured scaling and fault tolerance.

What is not new. We propose no new estimator or interval method; the one methodological proposal, conformal crediting, reuses standard conformal machinery for a new decision. Conformalized quantile regression [9], its missing-data augmentation [8], modality dropout [10], model-assisted estimation [11], [12], and the effects of plot size, saturation, allometric choice, and root-to-shoot choice [13]–[16] are established. Hash chains, sealed test sets, and commit-reveal are decades old [17]–[19]. The contribution is a blind, standards-anchored evaluation that tests these tools together on Indonesian and Bornean plots, under the failure modes an MRV platform meets: clouds, small plots, spatial transfer, and the deduction rules of the registries.

II. Related Work

A. Allometry and Standards

Pantropical allometry links D, H, and ρ to aboveground biomass (AGB) [1]; the BIOMASS package propagates its uncertainty [20]. Verra VM0047 and the HCSA toolkit specify sampling, measurement thresholds, and plot designs [2], [21]. Investigations of REDD+ crediting show that unverifiable baselines produce over-crediting [22], which motivates auditable estimates.

B. Remote Sensing of Biomass

Optical reflectance saturates over dense canopies. C-band radar mostly sees the upper canopy; L-band radar penetrates further and tracks woody biomass up to moderate densities [23], but its backscatter can *decrease* in very dense forest [24]. Spaceborne lidar from GEDI samples canopy height in 25 m footprints and is converted to biomass by footprint models calibrated on field data [3], [25]. Global AGB maps combine these signals: ESA CCI Biomass [4], the 2010 harmonized map of Spawn *et al.* [5], and the annual Chloris product [26]. Canopy height alone predicts plot carbon well where lidar is available [27].

C. Machine Learning and Honest Validation

Gradient-boosted trees are strong tabular regressors [28]; models fitted on log biomass need a retransformation correction [29]. Spatial autocorrelation makes random cross-validation optimistic; blocked or leave-region-out designs are required [6], [7]. Conformal prediction yields distribution-free intervals with finite-sample coverage [30]. Agreement is assessed with RMSE, bias, and Lin's concordance [31].

D. Clouds and Uncertainty

Southeast Asia is cloudy on at least 80% of days [32], and scene metadata suggest at most one cloud-free Sentinel-2 scene per year over much of tropical Indonesia [33]. The combined Sentinel-2 and Landsat revisit is 2.9 days [34]. Cloud masks such as Fmask and Cloud Score+ reach about 90% accuracy, with thin cloud and shadow the main failures [35]–[37]. Flores-Anderson *et al.* map the frequency of clear 30 m observations across the tropics [38]. Workarounds include best-available-pixel compositing [39], harmonized Landsat–Sentinel data [40], SAR–optical fusion [41], and deep SAR-to-optical cloud removal [42]. On the uncertainty side, split conformal prediction [43], conformalized quantile regression [9], normalized [44] and Mondrian [45] variants, and weighted or spatial conformal prediction under shift [46], [47] give distribution-free intervals; Zaffran *et al.* extend them to missing covariates by augmenting the calibration set with the test point's missingness pattern [8]. Conformal intervals have been applied to forest-attribute maps from Sentinel-2 [48] and to Earth-observation models in general [49]. For biomass maps, the CEOS protocol [50] and Duncanson *et al.* [51] recommend plots of at least 1 ha and explicit error propagation, and Araza *et al.* separate map error from plot error [14]. Model-assisted and model-based estimators that combine remote sensing with field samples are standard in forest inventory [11], [12], [52]; their efficiency grows with plot size [53], and model-based variance estimators under-cover when residual spatial correlation or model bias is ignored [54], [55].

TABLE I
Right-Grid Field Datasets (Stock)

Source	Region	Plots	Blocks	Date basis
HCSA [60]	Indonesia (6 prov.)	148	6	release 2023
GEO-TREES [61]	Pantropical	142	15	census year
Labrière [62]	Fr. Guiana, Gabon	186	6	campaign year
Total		**476**	**27**	

E. Integrity and Distributed Processing

Hash-chained time-stamping makes a document's existence at a time verifiable [17]; commit-reveal protocols bind a party to a value before another party reveals theirs [18]. Large geospatial workloads are processed as independent map tasks [56]; consistent hashing places keys on workers with minimal remapping [57], and Graham's longest-processing-time rule bounds makespan within $4/3$ of optimal [58]. Cloud catalogues such as the Microsoft Planetary Computer [59] and Google Earth Engine expose these archives through APIs.

III. Methodology

A. Right Grid: Field Ground Truth by the Standards

The right grid is the number a carbon standard would accept. We therefore build it the way the standards prescribe, from sampling design to deduction, and keep the deterministic point value separate from its uncertainty. Table I lists the plot sources and Table II the frameworks we follow.

What a ground-truth plot is. The coordinates of a plot only say where it is; the ground truth is the list of trees measured inside it. A field crew navigates to the coordinates, lays out the plot around the centre (for HCSA a 0.05 ha circle with a 0.01 ha inner circle for small stems), and tags every stem above the minimum diameter. For each stem it measures the diameter D at 1.3 m with a diameter tape and the height H with a clinometer or laser hypsometer, and it records the local name, which a botanist resolves to species, genus, and family. Wood density ρ, the oven-dry mass of wood per green volume, is a property of the species and is looked up in a wood density database by species, falling back to the genus or family mean. It matters because two stems of the same size can differ greatly in mass: in HCSA, *Campnosperma auriculata* has $\rho = 0.44$ and *Combretocarpus rotundatus* $0.70\,\mathrm{g\,cm^{-3}}$, so the second holds 1.6 times the biomass at the same D and H. HCSA publishes the full list: 2,528 stem records of 370 species in 89 families, each with D, H, local and scientific name, family, and the ρ assigned by the data provider from the ICRAF wood density database [60]. GEO-TREES and Labrière publish plot biomass that the networks computed from their own stem lists. None of these lists reaches the left grid, which receives only the manifest.

TABLE II
FIELD-INVENTORY RULES OF THE FRAMEWORKS APPLIED TO THE RIGHT GRID

Framework	Plot design (minimum stem size)	Confidence	Precision	Uncertainty rule
SNI 7724:2011 [63]	nested 2×2 (seedlings, litter), 5×5 (saplings 2–10 cm), 10×10 (poles 10–20 cm), 20×20 m (trees ≥20 cm); five pools, Tier 3	not stated	sampling error ≤20%	stratified systematic or random sampling
Indonesian NFI [64]	20 km grid; 100×100 m permanent plot of 16 units of 25×25 m; subplot radii 1/2/5 m; trees ≥20 cm	–	–	national FRL, Manuri allometry, CF 0.47
HCSA Toolkit v2.0 [21]	circular, $r = 5.64$ m for 5–14.9 cm, $r = 12.61$ m (0.05 ha) for ≥15 cm	90%	±10%	classes HDF, MDF, LDF, YRF, S, OL
IPCC 2006/2019 [65]–[67]	five pools; CF 0.47; root-to-shoot R by zone	95%	–	Approach 1 propagation or Approach 2 Monte Carlo
CDM AR-TOOL14/03 [68], [69]	stratified random plots	90%	±10%	stepped discount above 10%; sampling error only
Verra VM0047 v1.1 [2]	plot sampling, remeasured ≤5 yr	90%	–	deduct $U - 10\%$; zero credit if $U > 100\%$
Gold Standard A/R [70]	Winrock sourcebook plots; CF 0.475	90%	±20%	deduct the excess over 20%

Step 1, tree. Aboveground biomass of a tree with diameter D (cm), height H (m), and wood density ρ (g cm^{-3}) follows Chave 2014 Eq. 4 [1],

$$\mathrm{AGB}_i = 0.0673\,(\rho_i D_i^2 H_i)^{0.976}\ [\mathrm{kg}], \qquad \sigma_{\ln} = 0.357, \quad (1)$$

where $\sigma_{\ln}$ is the residual standard error on the log scale. Where height is not measured, a local model $\ln H = a + b \ln D$ with Duan correction replaces it (SAFE: $a = 0.549$, $b = 0.656$, residual SD 0.348); Chave Eq. 7 with the environmental stress term E is the fallback where no local heights exist. For HCSA, wood density is the per-stem value of the archive, and the level at which the Global Wood Density Database [71], [72] resolves each stem (species, genus, or family) sets its Monte Carlo error. For SAFE, wood density is the database mean at the finest level available, preferring South-East Asian records. As a check on model choice we also apply the East-Kalimantan equation of Basuki *et al.* [73], $\ln \mathrm{AGB} = -0.744 + 2.188 \ln D + 0.832 \ln \rho$ (correction factor 1.047). Palms are excluded: Chave's equations are for dicotyledonous trees, and palm allometry depends on stem height, not diameter [74], [75].

Step 2, plot. Tree biomass is expanded by the area in which a stem of its size is counted. For the HCSA nested design [21],

$$\mathrm{AGB}_j = \sum_{i \in j} \frac{\mathrm{AGB}_i}{1000\, a(D_i)}, \quad a(D) = \begin{cases} 0.01, & 5 \le D < 15 \\ 0.05, & D \ge 15 \end{cases} \quad (2)$$

in Mg ha^{-1}; SNI 7724 uses the same form, $C_n = C_x/1000 \times 10{,}000/l_{\mathrm{plot}}$, with its own subplot areas [63].

Step 3, pools and carbon. The plot carbon density is

$$C_j = \mathrm{CF}\,(1+R)\,\mathrm{AGB}_j + C_{\mathrm{DW},j} + C_{\mathrm{L},j} + C_{\mathrm{SOC},j}, \qquad \mathrm{CO_2e}_j = \tfrac{44}{12}\,C_j, \quad (3)$$

with CF $= 0.47$ (IPCC 2006 Table 4.3, unchanged in 2019; SNI 7724; Verra) or 0.475 (Gold Standard). For the root-to-shoot ratio R the frameworks disagree: SNI 7724 Lampiran A adopts the IPCC 2006 tropical-rainforest value $R = 0.37$, while the 2019 Refinement gives 0.207 (SD 0.072) below 125 Mg ha^{-1} and 0.212 (SD 0.077) above for natural Asian tropical rainforest [65], [66]. Dead wood, litter, and soil carbon are not in the public records; omitting a pool can only lower a stock claim, which is the conservative direction. All comparisons with the left grid use AGB, which removes the R and CF choices from the benchmark; we report their effect separately.

Step 4, stratum and project. Following AR-TOOL14 Eqs. 14–17 [68], plots are grouped into strata h (the HCS classes) with weights w_h, and the project mean and its relative uncertainty at 90% are

$$\bar{b} = \sum_h w_h \bar{b}_h, \qquad u = \frac{t_{0.95,\,n-M}\sqrt{\sum_h w_h^2 s_h^2 / n_h}}{\bar{b}}, \quad (4)$$

with s_h^2 the within-stratum variance, n_h plots in stratum h, n plots in total, and M strata. The mapped stratum areas are not public, so the plot shares stand in for w_h. The number of plots needed for a margin E (AR-TOOL03 Eq. 2, Neyman allocation, iterated on t when $n < 30$) [69] is

$$n = \left(\frac{t_{0.95}}{E}\right)^2 \left(\sum_h w_h s_h\right)^2, \qquad n_h = n\,\frac{w_h s_h}{\sum_h w_h s_h}. \quad (5)$$

Step 5, uncertainty. AR-TOOL14 counts sampling error only. The IPCC asks for all sources, combined by Approach 1, $U = \sqrt{\sum_k U_k^2}$ for products and $U = \sqrt{\sum_k (U_k x_k)^2}/|\sum_k x_k|$ for sums, or by Monte Carlo (Approach 2) [67]. We use Approach 2 with 2,000 draws per tree and the error models of the BIOMASS package [20]: diameter error with SD $0.0062D + 0.0904$ cm for 95% of stems and 4.64 cm for a random 5% [76]; height error of 10% [76]; wood-density SD of 0.0708, 0.0941, and 0.1234 g cm^{-3} at species, genus, and family level; the allometric residual $\varepsilon_i \sim \mathcal{N}(0, \sigma_{\ln}^2)$; CF $\sim \mathcal{N}(0.4713, 0.0206^2)$ [77]; and R with its 2019 SD. The project error budget splits the variance of $\bar{b}$ into a sampling part (Eq. 4) and a measurement-plus-allometry part (the Monte Carlo variance of the same weighted mean).

Step 6, change and deduction. Sequestration uses the IPCC stock-difference method, $\Delta C = (C_{t_2} - C_{t_1})/(t_2 - t_1)$ (Vol. 4

Eq. 2.5), on the same plots at both dates [65]. Its uncertainty follows VM0047 v1.1 Eq. 28 [2],

$$U = \frac{T\sqrt{SE_0^2 + SE_1^2 - 2\rho\, SE_0 SE_1}}{\Delta C}, \qquad \mathrm{UNC} = \min\big(1, \max(0,\ U - 0.10)\big), \tag{6}$$

where SE_0 and SE_1 are the standard errors of the mean stock at the two dates, ρ is their correlation on permanent plots, and T is the two-tailed t value at $\alpha = 0.1$. Credits are multiplied by $1 - \mathrm{UNC}$. For stocks, we report the deductions of VM0047 ($u - 10\%$), AR-TOOL14 (a stepped share of u: 0 up to 10%, then 25, 50, 75, 100% of u at 15, 20, 30%), Gold Standard ($u - 20\%$), and VMD0017 (95% uncertainty minus 15%) [68], [70], [78]. In the Monte Carlo for change, a tree keeps the same allometric and height residual at both censuses, because the same stem is measured with the same equation; we also run the case in which they are redrawn.

Data screening. For HCSA, Step 1 is recomputed for every tree. Sixty-two records stated an area that contradicts Eq. 2 and were expanded by the protocol area. Whole plots were excluded, never imputed, if any record had a missing input, $H < 1.3$ m, conflicting coordinates, or $D > 200$ cm. The last rule caught plot RU09 (Riau), where a *Gluta renghas* was recorded at $D = 480$ cm, which matches a circumference typed into the diameter column ($480/\pi \approx 153$ cm); left in place it would have produced 4,426 Mg ha^{-1}. Of 185 plots, 148 were retained; the 15 records labelled as oil palm all fell in excluded plots. GEO-TREES publishes plot AGB from forest networks [61]; 154 plots with withheld AGB were dropped. Labrière *et al.* publish 1 ha units with AGB from Chave Eq. 4 and local height-diameter models, plus 95% credible intervals [62]; we use the 1 ha units only.

The outcome table is written to a sealed file. Only a manifest of plot id, region block, coordinates, date, date precision, and extraction radius is released to the left grid.

B. Temporal Matching

Field inventories record a date or, more often, a year; no public inventory records the minute of measurement, and optical satellites revisit every 5–16 days with frequent cloud. We therefore match by window: for a plot measured at t_j, the left grid uses only clear observations in $[t_j - W,\ t_j + W]$, records the offset in days of the closest one, and takes the median of up to three clear scenes. The default is $W = 90$ days; RQ1 varies $W \in \{30, 90, 180, 365\}$ on the Indonesian plots. Each record also carries its timestamp precision (*day*, *year*, or *campaign year*) so that the analysis can separate sensor noise from date uncertainty.

C. Left Grid: Satellite Features

For each plot the left grid reads, inside a square window around the plot centre: Sentinel-2 L2A reflectance in ten bands with scene-classification cloud masking and seven indices (NDVI, EVI, NDRE, NDMI, NBR, IRECI, SAVI); Landsat 5–9 Collection 2 surface reflectance with QA cloud masking; Sentinel-1 RTC backscatter (VV, VH, ratio); the ALOS PALSAR annual L-band mosaic (HH, HV); and Copernicus 30 m elevation and slope. Two global products are read as zero-shot baselines: Chloris annual AGB (2003–2019) [26] and the 2010 harmonized map [5]. Through Google Earth Engine the left grid also reads the Meta 1 m [79] and ETH 10 m [80] canopy height maps, in the plot and in a 50 m buffer. It also reads GEDI L4A footprint biomass and L2A relative height within 500 m and 1 km (quality-filtered, April 2019 to March 2023), the GEDI L4B 1 km grid [25], [81], ESA CCI Biomass for the nearest year [4], the ALOS/ALOS-2 PALSAR mosaic for the nearest year (2007–2024), and Hansen forest loss up to the field year [82]. The canopy height maps describe 2018–2020 imagery and GEDI 2019–2023, so for plots measured earlier they are not time-matched.

Algorithm 1 Satellite to Carbon for One Plot (E2)

Require: plot coordinates, field date t, radius r (the manifest)
for each sensor (Sentinel-2, Landsat, Sentinel-1, PALSAR) **do**
  find scenes within $t \pm 90$ days
  keep a scene only if every pixel of the plot window is clear
  window means of bands; indices such as NDVI (Eq. 7)
  median over up to three clear scenes
end for
add terrain, canopy height percentiles, GEDI biomass and height
$\hat{y} \leftarrow f_{-R}(x)$, LightGBM on ln AGB trained on the plots of every other region
$\widehat{\mathrm{AGB}} \leftarrow e^{\hat{y}} \cdot \overline{e^{\varepsilon}}$ (Duan smearing)
return $C = 0.47\,\widehat{\mathrm{AGB}}$, $\mathrm{CO_2e} = \frac{44}{12} C$ per hectare

From pixels to carbon. A satellite does not see trees. It records, per pixel, the light reflected in each band and the radar echo; GEDI records the height profile of 25 m footprints. Optical bands are combined into vegetation indices, of which the best known is the normalized difference vegetation index [83],

$$\mathrm{NDVI} = \frac{R_{\mathrm{NIR}} - R_{\mathrm{red}}}{R_{\mathrm{NIR}} + R_{\mathrm{red}}}, \tag{7}$$

with R the surface reflectance in Sentinel-2 band B8 (842 nm) and B4 (665 nm), or Landsat bands 4 and 3 (TM) and 5 and 4 (OLI). Chlorophyll absorbs red light for photosynthesis and the cell structure of leaves reflects near-infrared, so a green canopy reads about 0.8, bare soil 0.1–0.2, and water below zero. NDVI measures how green a pixel is, not how much wood stands under the leaves; Section IV-E shows that it saturates in closed forest. Algorithm 1 gives the whole chain from pixels to carbon for one plot.

D. Clouds

Southeast Asia is cloudy on at least 80% of days, with no reliable clear season [32], and Sentinel-2 metadata for 2017 showed at most one cloud-free scene over much of tropical Indonesia [33]. Tropical maps of clear-observation frequency exist at 30 m [38], but we found no Sentinel-2 statistic at Indonesian plot scale and no measurement of how often scene-level filters misjudge a plot, so we measured both, then designed the estimator so that clouds cost as little as possible.

C1, plot-level climatology. For 97 locations (one per 0.1° cell holding a plot), we read the Sentinel-2 scene classification (SCL) of every acquisition from 2021 to 2023 in a 5×5 window of 20 m pixels around the plot: 22,697 acquisitions after duplicates of the same date were merged. An acquisition is *clear* if every pixel is vegetation, bare soil, water, or unclassified. From this we estimate the per-acquisition clear probability p, its monthly cycle, the gaps between clear looks, and the probability of at least one clear look within $\pm W$ days of a random date, which we compare with the independence model $1-(1-p)^m$ for m acquisitions in the window; the difference measures how much clouds cluster in time. We also test the scene-level filter used by many Earth Engine pipelines, including our own platform (`CLOUDY_PIXEL_PERCENTAGE` <30 or <40), against the plot condition.

C2, missing optical data. When clouds hide a plot, the optical block (Sentinel-2 and Landsat features) is missing. For a fraction $q \in \{0, 0.25, 0.5, 0.75, 1\}$ of each target region's plots we remove it and compare four strategies, all leave-one-region-out: *naive* (the full model, missing values routed down LightGBM's default branches); *dropout*, one model trained on the real rows plus copies with the optical block removed, the modality-dropout idea of Neverova *et al.* [10]; *switch*, a cloud-aware ensemble that uses a radar-and-terrain model when optical data are missing; and *impute*, optical features regressed from radar, a simple form of SAR-to-optical gap filling [42].

Cloud-aware conformal intervals. A prediction interval is only honest if it is calibrated on cases like the one being predicted. When clouds are rare in the calibration data, a group-conditional (Mondrian) calibration [45] of the cloud-hit group has too few scores. We therefore use the missing-data augmentation of Zaffran *et al.* [8]: every inner out-of-fold calibration plot is scored twice, as observed and with its optical block removed, with quantile models trained under modality dropout [10], and each test plot takes the threshold of its own missingness pattern. With two patterns this is a special case of their CP-MDA. Their guarantee assumes exchangeable data and missingness independent of the outcome; here clouds follow climate and region, and calibration is out-of-region, so neither holds and coverage has to be measured. We test it on every target plot, both as observed and fully clouded.

E. Estimator Ladder

E0 (off the shelf). Global products read at the plot, no fitting. **E1 (no field data).** The published GEDI L4A and L4B biomass products, read as they are, serve as the estimator that uses no field plot of ours. A model trained on GEDI footprint labels is implemented but not evaluated. **E2 (transfer).** LightGBM on ln AGB, trained on all region blocks except the target block and predicting it blind (leave-one-region-out), with Duan smearing [29]. **E3 (k-shot).** E2 followed by a ratio calibration on k randomly drawn plots of the target block, scored on its remaining plots, repeated 200 times per k. Split-conformal intervals for E2 are calibrated on inner leave-one-region-out residuals of the training blocks only [30].

Algorithm 2 E4: Count the Trees, Then the Field Formula

Require: 1 m canopy height patch h, plot footprint, realm, region
$h \leftarrow$ Gaussian smoothing of h, $\sigma = 1$ m
tops $\leftarrow \{p : h(p) \geq 5,\ h(p) = \max_{|q-p| \leq 0.06h(p)+0.5} h(q)\}$
grow crowns from the tops, highest pixel first, while $h \geq \max(2,\ h_{\text{top}}/2)$ and within 15 m
for each top i inside the footprint **do**
    $H_i \leftarrow h(\text{top}_i)$, $CD_i \leftarrow 2\sqrt{A_i/\pi}$ (A_i crown area)
    $D_i \leftarrow a(H_i CD_i)^b$ (Eq. 8); drop the tree if $D_i < 5$ cm
    $\text{AGB}_i \leftarrow 0.0673(\bar{\rho}\, D_i^2 H_i)^{0.976}$, $\bar{\rho}$ the regional default
end for
return $\sum_i \text{AGB}_i/(1000\, A_{\text{plot}})$ in Mg ha^{-1}, then $\times 0.47 \times \frac{44}{12}$

E4 (count the trees, then the field formula). E2 learns a direct map from pixels to biomass. E4 instead rebuilds the right grid's own tree list from the 1 m canopy height map and applies the right grid's formula to it, so that the two grids differ only in where each tree's numbers come from (Algorithm 2). A 140×140 m patch of the Meta map is read on the local UTM grid at 1 m around each plot. The map stores whole metres, so it is smoothed by a Gaussian of 1 m. Tree tops are local maxima taller than 5 m in a window whose radius grows with height, $r = 0.06H + 0.5$ m, the variable window filter of Popescu and Wynne [84]. Each crown is flooded from its top, highest pixels first, while the canopy stays at least 2 m tall, at least half the top's height, and within 15 m. The stem diameter follows from height and crown diameter by Jucker *et al.* [85],

$$D = a\,(H \cdot CD)^b \text{ [cm]}, \tag{8}$$

with the tropical-angiosperm coefficients of the plot's realm: $(a, b) = (0.486, 0.802)$ for Indo-Malaya, $(0.428, 0.821)$ for the Afrotropic, and $(0.591, 0.781)$ for the Neotropic. A satellite cannot identify species, so every tree takes the mean wood density of the tropical species of its region in the Global Wood Density Database (each species counted once): 0.574 for South-East Asia, 0.615 for Africa, and 0.630 for South America. Eq. 1 then gives each tree's biomass, the trees whose top falls inside the plot footprint (0.05 ha for HCSA, 1 ha otherwise) are summed, and the sum is divided by the footprint area. E4 uses no field data at all. *E4 scaled* multiplies it by the ratio of field to E4 totals on all other regions, which is the least field data a correction can use. E4 was designed after the stock reveal and is exploratory; as a sensitivity we also run it without smoothing.

F. Confidence: Plot and Project Intervals

Plot intervals. All intervals target 90% and are calibrated on inner leave-one-region-out residuals of the training regions only. With calibration scores $s_1, \ldots, s_n$, the conformal threshold is the $\lceil (n+1)(1-\alpha) \rceil$-th smallest score, which guarantees marginal coverage between $1-\alpha$ and $1-\alpha+1/(n+1)$ under

exchangeability [43]. We compare four scores on $y = \ln \mathrm{AGB}$:

$$\text{split:} \quad s = |y - \hat{y}|, \qquad \hat{C} = \hat{y} \pm q, \tag{9}$$

$$\text{Mondrian:} \quad \text{split, one } q_g \text{ per availability group } g, \tag{10}$$

$$\text{normalized:} \quad s = |y - \hat{y}|/\hat{\sigma}(x), \qquad \hat{C} = \hat{y} \pm q\,\hat{\sigma}(x), \tag{11}$$

$$\text{CQR:} \quad s = \max\{\hat{q}_{\text{lo}}(x) - y,\ y - \hat{q}_{\text{hi}}(x)\}, \tag{12}$$

$$\hat{C} = [\hat{q}_{\text{lo}}(x) - q,\ \hat{q}_{\text{hi}}(x) + q], \tag{13}$$

following Papadopoulos *et al.* [44] for the normalized score, with $\hat{\sigma}(x)$ a second LightGBM fitted to the inner residuals, and Romano *et al.* [9] for conformalized quantile regression (CQR), with LightGBM quantile models at 5% and 95%. We report coverage, the median ratio of the upper to the lower bound, and the Winkler interval score on the log scale, $(\hat{u} - \hat{l}) + \frac{2}{\alpha}\big[(\hat{l} - y)_+ + (y - \hat{u})_+\big]$. As Araza *et al.* note, an interval calibrated on plots covers the plot value, which includes field error, not the true stock [14].

Is a metadata "confidence score" a confidence? Monitoring platforms often grade their output with a score built from cloud cover, data completeness, and temporal offset. Ours did: 40 points for cloud cover, plus weights for completeness and stability. We rebuild such a score from the same inputs (40 points for clear optical scenes, 30 for the share of sensor groups present, 30 for temporal closeness) and test whether it ranks plots by their actual error, using Spearman's ρ against $|\ln y - \ln \hat{y}|$, next to the learned $\hat{\sigma}(x)$, both raw and as a partial rank correlation that removes predicted AGB and the dataset or region, which could otherwise confound it.

Project intervals. Every region block with at least eight plots plays a project, whose true stock is the mean field AGB of its plots. For satellite-only estimates of the mean, a naive interval treats plot errors as independent, $SE = s_w/\sqrt{n}$. Map errors, however, are spatially correlated over 50–100 km [14], so a project shares one regional bias. The two-component interval adds the between-region variance of the region-mean log residual,

$$SE^2 = \frac{s_w^2}{n} + s_b^2, \tag{14}$$

with s_w and s_b estimated from the other regions only. When k field plots are available, the satellite becomes auxiliary information for a model-assisted estimator [11], [52]. The ratio estimator is

$$\hat{\bar{Y}}_R = \frac{\sum_{i \in s} y_i}{\sum_{i \in s} \hat{y}_i}\,\bar{\hat{y}}, \qquad \widehat{V} = \Big(1 - \frac{k}{n}\Big)\frac{1}{k}\,s_e^2, \quad e_i = y_i - \hat{R}\hat{y}_i, \tag{15}$$

where $\bar{\hat{y}}$ is the satellite mean over the whole project. The regression (GREG) estimator is $\bar{y}_s + \hat{\beta}(\bar{\hat{y}} - \bar{\hat{y}}_s)$. Both are compared with the field-only mean of the same k plots over 500 random draws. The relative efficiency $\mathrm{MSE}_{\text{field}}/\mathrm{MSE}_{\text{estimator}}$ says how many field plots one satellite-assisted plot is worth. We use the empirical MSE over draws, because the GREG variance formula underestimates at small k. A conservative estimate $\hat{Y}(1 - \mathrm{UNC})$ with the VM0047 rule exceeds the truth whenever the project is over-credited; we report how often.

G. Conformal Crediting

A registry does not issue an interval; it issues a number. The number should be a lower bound that exceeds the true stock in at most a stated share δ of projects. This is the conservativeness principle of Grassi *et al.* [86], which Neeff writes as $P\{\hat{Y}(1-d) \le Y\} \ge 1 - \delta$ and solves parametrically with $d = t\,SE/\hat{Y}$ [87]. A parametric bound is only as good as its standard error. For a satellite map no design-based standard error exists, and the model-based one rests on an error model we have shown to be fragile. We therefore calibrate the bound on other projects whose field truth is known, as Singh *et al.* suggested in passing for registry intervals [49].

Rule. Let project j have an estimate $\hat{Y}_j$ of its mean stock, true mean Y_j, and a relative standard error r_j ($r_j = 1$ when none exists). Its score is $z_j = \ln(\hat{Y}_j/Y_j)/r_j$. For a new project, with scores $z_1, \ldots, z_m$ from m calibration projects, the credited stock is

$$C = \hat{Y}\exp(-q\,r), \qquad q = z_{(\lceil (m+1)(1-\delta) \rceil)}, \tag{16}$$

the $\lceil (m+1)(1-\delta) \rceil$-th smallest score, or $C = 0$ if that rank exceeds m.

Guarantee. If the new project and the calibration projects are exchangeable, then $P(C > Y) \le \delta$, and it is at least $\delta - 1/(m+1)$ when scores have no ties. This is split conformal prediction [43] with projects as the units. Plots inside a project may be correlated in any way, because each project contributes one score [88]. The bound needs $m \ge 1/\delta - 1$ calibration projects: 9 for $\delta = 0.1$ and 19 for $\delta = 0.05$.

Invariance to map inflation. If every satellite estimate is multiplied by $c > 0$, every plain score ($r = 1$) shifts by $\ln c$, so q shifts by $\ln c$, and $C = c\hat{Y}\exp(-q - \ln c) = \hat{Y}\exp(-q)$ is unchanged. A developer who inflates a map uniformly gains no credit. A parametric bound $\hat{Y}\exp(-z_{1-\delta}SE)$ with SE from residual spread grows by the full factor c.

Variants and baselines. The *studentized* variant uses the design-based relative standard error of the project estimate as r_j. With k field plots this is the ratio estimator of Eq. 15, so the variant recalibrates the registry variance formula across projects. Breidenbach *et al.* found that such formulas undercover [54]. The *Mondrian* variant calibrates within the plot-size class, small (radius $\le$15 m) or large, which is a public field of the manifest. Baselines are no deduction; the VM0047 deduction $\hat{Y}(1 - \max(0, U - 0.1))$; the design lower bound $\hat{Y}(1 - t_{1-\delta,k-1}r)$ of Neeff [87]; and the parametric satellite-only bounds with independent or two-component errors (Eq. 14).

Integrity-constrained yield. We score a rule by its realised over-crediting rate and by its *yield*, the mean of C_j/Y_j: the share of the true stock that is credited. Comparing designs by yield at a fixed realised risk gives the value of a field plot in credits.

Experiment. Projects are region blocks with at least 8 plots. Regions of 16 or more plots are split into spatial sub-projects of about 10 plots by k-means on coordinates, which gives 31 projects in 19 regions. A project is calibrated only on projects

from other regions, so the guarantee is tested under transfer. With k field plots, each project is drawn 200 times. Each draw is calibrated on one random draw per calibration project (the subsampling scheme for two-layer data [88]). Every run comes after the stock reveal, so the results are exploratory.

H. Scoring

We report RMSE, relative RMSE, bias, R^2, and Lin's concordance [31] per dataset and overall, interval coverage for E2, error by field-AGB range, error versus window W, and the error of the mean of n plots as n grows, which approximates estimating a larger area.

I. Sequestration

Credits are issued on change, not stock. The only public tropical dataset we found that gives repeated per-stem measurements with *day-precision* dates is the SAFE Project tree census in Sabah, Malaysian Borneo [89], [90]: $25\,\mathrm{m}\times25\,\mathrm{m}$ plots across old-growth, logged, riparian, and oil-palm land, censused from 2010 to 2020, with every stem's measurement date recorded. It is also the closest field site to Kalimantan in our benchmark.

The right grid applies the same chain to every census: Chave Eq. 4 on stems with $D \geq 10\,\mathrm{cm}$; height from a local $\ln H = a + b \ln D$ model fitted to the heights SAFE measured, with Duan correction; wood density from the Global Wood Density Database [71], [72] by species, then genus, then family, then plot mean, preferring South-East Asian records. For each plot with censuses at least three years apart, sequestration is

$$\Delta\mathrm{AGB}_j = \frac{\mathrm{AGB}_j(t_1) - \mathrm{AGB}_j(t_0)}{t_1 - t_0} \quad [\mathrm{Mg\,ha^{-1}\,yr^{-1}}], \quad (17)$$

where t_0 and t_1 are the median stem dates of the first and last census. This gives 223 plots in 18 blocks, with t_0 between 2010 and 2015 and t_1 between 2016 and 2020. The manifest released to the left grid holds only plot id, block, coordinates, and the two dates.

The left grid estimates ΔAGB three ways: *no change* ($\Delta = 0$, the baseline any estimator must beat); *S0*, the difference of the Chloris annual maps between the census years; and *S2*, the difference of two E2 stock predictions at t_0 and t_1, with E2 trained on the 476 stock plots only. Sentinel-2 did not exist at most t_0 dates, so S2 uses Landsat, Sentinel-1, PALSAR, and terrain. No SAFE label is released to any estimator; the same E2 model also predicts SAFE stock at t_0, which is an out-of-region test on Borneo.

J. Blindness Ledger

Every protocol step is an entry in an append-only log whose entries carry the SHA-256 of the previous entry [17]. An evaluation epoch is: `seal` (digest of the sealed outcomes and of the public manifest), `commit` (digest of each estimator's prediction file), `release` (for E2, the digest and size of each fold's training labels, with a flag stating whether any target-block label was included), and `reveal` (accepted only if the sealed file still matches its digest). Commits after a reveal are refused. Editing any entry breaks every later hash, so a third party can verify the order of events from the log alone [18].

K. Distributed Extraction

One task is one (plot, date, window, sensor) request. Its result is stored under the SHA-256 of the canonical task, which makes re-execution idempotent and turns the cache into a checkpoint: a restarted run skips every finished task. Writes are atomic (write then rename). Transient errors are retried with capped exponential backoff and jitter. Tasks are grouped into spatial tiles so a worker reuses scenes for neighbouring plots, but field plots cluster (one site holds 89 plots), so a tile holding more than the per-worker mean is split into sub-buckets by plot id, keeping all sensors of a plot together. Buckets are then placed either by consistent hashing [57] or by the longest-processing-time rule [58]. On the 2,052 tasks of this study with 24 workers, the worst-worker load factor $\lambda = \max_w L_w / \bar{L}$ was 6.75 for tile-only placement, 2.18 for split tiles with consistent hashing, and 1.40 for split tiles with LPT.

IV. Results

A. How Certain Is the Right Grid?

Before scoring the satellites we score the field answer.

Plot level. On the 148 HCSA plots, the Monte Carlo 95% interval of plot AGB has a median half-width of 31% (IQR 25–45%). Of the stems, 45% have a species-level wood density, 42% genus-level, 7% family-level, and 5% none. The Monte Carlo median is 5% above the deterministic value, because the allometric error is log-normal; the deterministic value is the one the standards use.

Equation choice. Consistent with earlier comparisons of generic and regional equations [15], the regional equation of Basuki *et al.* gives almost the same total over all plots (ratio 1.01). By region, however, the stock moves from -40% (Central Kalimantan) to $+121\%$ (South Sumatra), because it ignores height where HCSA measured short trees.

Pool choice. The root-to-shoot ratio of SNI 7724 (IPCC 2006) gives 13% more total carbon than the 2019 Refinement. This is close to the 12% shift in global vegetation carbon that Mokany *et al.* found from revising root-to-shoot ratios [16]. Two standards, both valid, differ by more than the precision target of most methodologies.

Project level. Table III applies AR-TOOL14 to each province. The sampling uncertainty at 90% is 9–34%, and five of six provinces meet SNI 7724's 20% limit. But measurement and allometric error, which AR-TOOL14 does not count, carry 10–64% of the variance of the project mean. Their share is largest where the plots vary least between each other (Riau, 64%). The deductions differ by standard for the same data (Jambi: 24% under VM0047, 34% under AR-TOOL14, 14% under Gold Standard). Reaching $\pm10\%$ at 90% would take 8 to 71 plots per province.

Change. Growth is far harder than stock. On SAFE, the Monte Carlo SD of a plot's ΔAGB is $2.0\,\mathrm{Mg\,ha^{-1}\,yr^{-1}}$ when a tree keeps its allometric residual across censuses, against

TABLE III
RIGHT-GRID UNCERTAINTY PER PROVINCE (HCSA, AGB)

Province	n	$\bar{b}$	u_{90} (%) Sampl.	u_{90} (%) Total	Meas. share	VM0047 deduct.	Plots for ±10%
C. Kalimantan	33	250	13.8	14.0	10%	3.8%	36
Jambi	20	171	33.9	34.5	16%	23.9%	71
Riau	19	126	12.7	19.8	64%	2.7%	19
S. Sumatra	22	80	9.2	12.3	51%	0.0%	14
W. Kalimantan	36	165	14.4	15.1	15%	4.4%	23
W. Papua	18	116	8.8	12.4	57%	0.0%	8

$\bar{b}$: stratified mean AGB (Mg ha^{-1}). Meas. share: measurement and allometry share of the variance of $\bar{b}$.

TABLE IV
BLIND STOCK ACCURACY (MG AGB HA^{-1}; COMMITTED BEFORE REVEAL)

Subset	Estimator	RMSE	rRMSE	Bias	R^2	CCC
All (476)	E0 Chloris	128.1	49.3%	−4.6%	0.29	0.43
	E0 HGB 2010	149.4	57.5%	−8.4%	0.03	0.27
	E2 LORO	126.1	48.5%	−9.5%	0.31	0.59
GEO-T. (142)	E0 Chloris	102.8	37.5%	−10.3%	0.37	0.56
	E2 LORO	114.9	41.9%	0.0%	0.21	0.55
HCSA (148)	E0 Chloris	172.4	107.1%	+19.0%	−0.12	0.02
	E2 LORO	165.6	102.9%	−29.1%	−0.03	0.22
Labr. (186)	E0 Chloris	101.2	31.0%	−10.3%	0.18	0.37
	E2 LORO	93.7	28.6%	−7.9%	0.30	0.55

4.3 when it is redrawn. The signal itself is 1.9. Remeasuring the same stems with the same equation cancels half the error. At block level, Eq. 6 gives 90% uncertainties of 101–901% when the correlation of the two stock means is ignored ($\rho = 0$), which is a 100% deduction for 13 of 15 blocks and over 90% for the other two. With the measured ρ (0.45–0.99), the uncertainty falls to 28–469%, and seven blocks keep some credit, the best at an 18% deduction. With 9–16 plots of 0.06 ha per block, even the field standard cannot certify growth of 2 Mg ha^{-1} yr^{-1}, unless it uses the correlation that permanent plots provide.

B. Blind Accuracy of Carbon Stock (RQ1)

Table IV gives the blind scores. All three estimators were committed to the ledger before the reveal. Over all 476 plots the best blind estimate, E2, misses the field value by a relative RMSE of 48.5% and explains 31% of its variance; the Chloris product, which was never fitted to any of our plots, is almost as close (49.3%). The 2010 harmonized map is worse (57.5%, $R^2 = 0.03$), partly because it describes 2010 while most plots were measured years later. Because carbon is a fixed multiple of AGB on both grids, every relative figure applies unchanged to t C ha^{-1} and t CO$_2$e ha^{-1}.

The dataset matters more than the estimator. On the 1 ha Labrière units E2 is within 28.6% (CCC 0.55); on GEO-TREES within 41.9%; on the Indonesian HCSA plots every estimator misses by more than 100% and none beats predicting the mean ($R^2 < 0$).

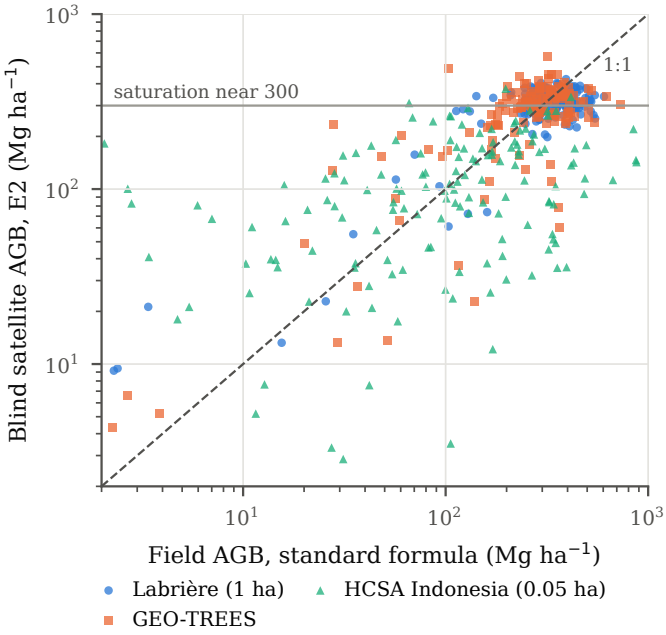


Fig. 2. Blind E2 prediction against the field value for 476 plots (log axes). Above about 300 Mg ha^{-1} predictions flatten; small HCSA plots scatter widely in both directions.

C. Where the Error Comes From

Saturation. Fig. 2 shows the pattern. Binned by field AGB, both E0 and E2 regress toward ∼300 Mg ha^{-1}. For plots above 400 Mg ha^{-1} (mean 480) the E2 mean prediction is 293 and the Chloris mean is 295, a median under-estimate of about one third. Below 50 Mg ha^{-1} (mean 21) E2 predicts 60 and Chloris 178. Between 100 and 300 Mg ha^{-1} the median error of E2 is only +5 to +12%. The satellite signal therefore carries useful information in the middle of the range and little at either end, which is the behaviour reported for optical and L-band signals [23], [24].

Scale. The HCSA plots are nested 0.01/0.05 ha plots with a median of 15 stems, so one large tree more or less moves the plot by hundreds of Mg ha^{-1}. Their field values span 2 to 876 Mg ha^{-1} within one province. A 30 m pixel window cannot resolve that, and the 4.6 km Chloris pixel averages open land and forest together, which explains its large over-estimate on HCSA open-land and scrub plots. The HCSA land-cover labels would separate these plots well (within-class standard deviations of 10–40 Mg ha^{-1} for all classes except high-density forest), but they are released with the field records, so they belong to the right grid and are never shown to the left grid. A satellite classifier that reproduced the HCS stratification is a promising left-grid estimator for future work.

Field truth is not the bottleneck. Labrière publish a 95% credible interval for every unit; the implied standard error of the field value is 5.0% (RMS), against a 28.6% satellite error on the same units. The right grid is far more precise than the left grid, so the gap is a satellite gap, not a ground-truth gap.

TABLE V
IMPROVEMENT LADDER, BLIND RRMSE (%)

Left-grid inputs	All	GEO-T.	HCSA	Labr.
Optical (S2 + Landsat)	56.9	51.2	103.0	42.0
Radar (S1 + PALSAR)	54.5	45.9	114.7	33.7
Optical + radar	52.1	46.4	103.7	33.9
+ terrain (= E2)	49.1	42.0	104.8	28.8
+ Chloris as feature	47.9	38.1	105.7	27.9
+ plot footprint	47.6	38.6	103.5	28.1
Mean of E2 and Chloris	**45.4**	**35.0**	**98.8**	28.0

TABLE VI
SHARPER INPUTS: BLIND RRMSE (%), LEAVE-ONE-REGION-OUT

Inputs	All	GEO-T.	HCSA	Labr.
E2 (S1/S2, Landsat, PALSAR, DEM)	49.1	42.0	104.8	28.8
+ ESA CCI	47.3	38.6	101.9	28.5
+ GEDI	45.5	36.6	102.4	24.5
+ canopy height	41.5	32.2	**94.8**	22.4
+ canopy height + GEDI	**40.1**	**27.7**	95.6	**20.6**
Height + GEDI only	43.2	31.7	103.1	20.7
Height power law (1 m)	50.8	35.0	115.5	30.7
GEDI L4A, as published	59.2	50.0	107.0	46.6
ESA CCI, as published	49.4	38.8	105.4	31.5
Gain of best over E2	-9.0	-14.3	-9.2	-8.2

Region-bootstrap 95% intervals of the gain: all -13.0 to -6.1; GEO-TREES -27.4 to -7.2; HCSA -13.2 to -2.9; Labrière -12.8 to -3.1.

Time. Re-extracting the Indonesian plots' optical and C-band features with $W = 30$, 90, 180, and 365 days changed their blind error by less than three points (rRMSE 102.5, 104.8, 102.7, 102.0%), while the share of plots with at least one clear Sentinel-2 scene rose from 74% at 30 days to 92% at 90 and 100% at 180. A tighter window costs coverage in the tropics and buys no accuracy when the field date is only known to the year, and when plot-scale noise dwarfs seasonal change. Temporal matching matters where the forest changes between the two dates, which is the sequestration case below.

D. Closing the Gap (RQ2)

Better models. Table V adds one information source at a time, always leave-one-region-out. These variants were run after the reveal, so they are exploratory; every variant we tried is reported. Radar alone is ahead of optical alone (54.5% vs 56.9% overall; 33.7% vs 42.0% on the 1 ha units), the two together beat either, terrain helps further, and feeding the Chloris product to the model as one more feature helps again. Of these steps, only the last one is clearly outside sampling noise (Section IV-J): the simple average of E2 and Chloris is the best blind estimator overall (45.4%, $R^2 = 0.39$), 3.7 points better than E2 with a region-bootstrap 95% interval of -7.2 to -0.6. None of these steps rescues the HCSA plots.

Sharper inputs: canopy height and spaceborne lidar. The largest model gain came from inputs, not algorithms. Through Earth Engine we added, for every plot, the Meta 1 m canopy height map [79] and the ETH 10 m map [80], read in and around the plot. We also added GEDI L4A footprint biomass and L2A height within 500 m and 1 km, together with the GEDI L4B grid [25], [81], and the ESA CCI Biomass map [4]. All of these are read from coordinates only. Table VI adds them to E2, always leave-one-region-out. Canopy height alone cut the overall error from 49.1% to 41.5%, and canopy height plus GEDI cut it to 40.1% (R^2 0.29 to 0.53, CCC 0.58 to 0.73). The gain is outside sampling noise in every dataset: overall it is -9.0 points, with a region-bootstrap interval of -13.0 to -6.1. On the 1 ha units the error fell from 28.8% to 20.6%, on GEO-TREES from 42.0% to 27.7%, and on HCSA from 104.8% to 95.6%. Used as published, without calibration, the GEDI products under-estimate by 29–41% and are worse than the raw map. They help as inputs, not as answers. A height-only power law fitted on the 1 m map (AGB $= aH^b$) is no better than E2, because one height-biomass curve does not transfer across regions. The improvement carries to project scale. With no field plot, the error of a block total fell from 13% to 7% on the 1 ha units, from 20% to 9% on GEO-TREES, and from 28% to 20% on HCSA (median over blocks). The mean of 20 plots is within 3–5% of its field mean outside Indonesia.

A few local plots. The largest gain comes not from the model but from field data in the target region. Fig. 3(b) shows E3: with k plots of the target block used for a ratio correction, the error of the block total falls from 29% (GEO-TREES) and 11% (Labrière) at $k = 0$ to 9–10% and 6% at $k = 10$–20 (median over blocks and 200 draws). On HCSA, calibration hurts until $k = 20$ (27% at $k = 0$, 53% at $k = 2$, 24% at $k = 20$): a small plot is so noisy that a handful of them is a worse calibrator than none. How many local plots are needed depends on plot size as much as on the satellite.

Aggregation. Credits are issued for project areas, not single plots. The error of the mean of n plots shrinks with n until only systematic bias remains (Fig. 3(a)). For the 1 ha units the median absolute error falls from 27% for one plot to 12% for ten and 11% for twenty; for HCSA from 188% to 27–30%. The floor of roughly 10% outside Indonesia and 25–30% in Indonesia is the regional bias that only local calibration removes.

Intervals are analysed together with clouds in Section IV-G.

E. Counting Trees from Space (RQ2)

What the field counts. The median HCSA plot holds 700 stems per hectare (200 of 15 cm or more) of five species. Table VII shows where its carbon is: stems of 5–15 cm are three quarters of the count but an eighth of the carbon, and the 6% of stems of 30 cm or more hold 58%. A method that finds the large trees finds most of the carbon; a method that counts stems is mostly counting saplings.

The wood. Table VIII groups every HCSA stem by the wood density of its species into light ($\rho < 0.5\,\mathrm{g\,cm^{-3}}$), medium (0.5–0.7), and heavy (≥ 0.7) wood. Light wood, mostly Lauraceae, Moraceae, and Anacardiaceae, is a quarter of the stems but a sixth of the carbon. Heavy wood, led by Dipterocarpaceae, Lauraceae, and Myrtaceae, is a third of the stems and two fifths of the carbon. Dipterocarpaceae appear in both the medium and the heavy class because the family spans both. The stem-weighted mean is $0.627\,\mathrm{g\,cm^{-3}}$, against

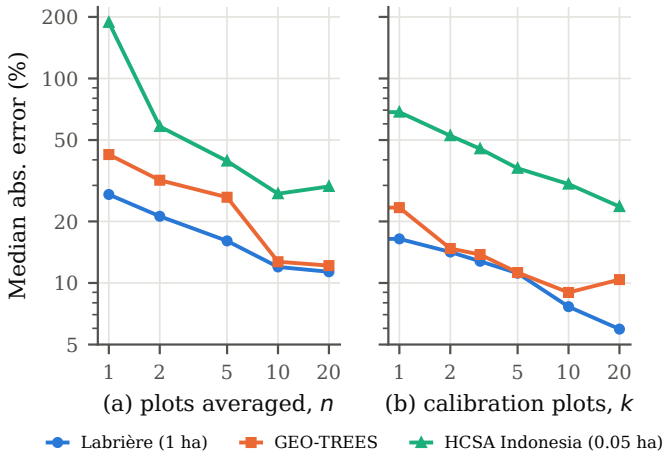


Fig. 3. Closing the gap. (a) Error of the mean of n plots against its field mean; (b) error of the block total after a ratio calibration on k local plots. Median over blocks and 200–300 random draws; log axes.

TABLE VII
HCSA Stems by Diameter: Count and Carbon

Diameter	Stems ha^{-1}	Stems (%)	Carbon (%)
5–15 cm	647	75.3	12.6
15–30 cm	157	18.3	29.1
30–50 cm	43	5.0	31.0
$\geq$ 50 cm	12	1.4	27.2

Stems per hectare averaged over the 148 plots, after the nested-area expansion of Eq. 2.

TABLE VIII
HCSA Stems by Wood Density Class

Class (g cm^{-3})	Spp.	Stems	C	$\bar{\rho}$	Leading families (by carbon)
light ($<$0.5)	83	24	16	0.46	Lauraceae, Moraceae, Anacardiaceae
medium (0.5–0.7)	179	44	44	0.60	Dipterocarpaceae, Moraceae, Anacardiaceae
heavy ($\geq$0.7)	127	32	40	0.78	Dipterocarpaceae, Lauraceae, Myrtaceae

Stems and C: share of stems per hectare and of carbon, %. One stem of *Myristica argentea* (plot PB01) is recorded at $\rho = 2.47$, which is physically impossible; it holds 1.1% of its plot and is left as published.

TABLE IX
Carbon Found by E4 and by the Map, Against the Field (t CO$_2$e ha^{-1} and %)

Subset (n)	Field	Estimate	Sat.	Total	Med.	$\pm$25%
All (475)	449	E4, as is	111	25	23	1
		E4, unsmoothed	178	41	37	4
		E4, scaled	427	97	89	36
		Map (E2 + CH + GEDI)	417	98	100	60
GEO-TREES (142)	473	E4, as is	115	24	24	0
		E4, unsmoothed	180	38	35	2
		E4, scaled	435	92	89	42
		Map (E2 + CH + GEDI)	467	99	99	72
HCSA (148)	277	E4, as is	64	23	7	3
		E4, unsmoothed	98	35	20	10
		E4, scaled	245	88	26	9
		Map (E2 + CH + GEDI)	199	72	90	22
Labrière (185)	567	E4, as is	147	26	26	0
		E4, unsmoothed	241	43	43	1
		E4, scaled	567	100	103	52
		Map (E2 + CH + GEDI)	554	98	101	81

Field and Sat.: mean t CO$_2$e ha^{-1}. Total: satellite carbon as % of field carbon over the plot footprints. Med.: median plot %. $\pm$25%: % of plots within 25% of the field value. CH: canopy height.

the 0.574 that E4 must assume. Giving every stem that default instead of its species value leaves 91% of the total carbon and 78–115% of a single plot's (10th to 90th percentile). Wood density alone, the one input a satellite cannot see at all, is therefore worth about a tenth of the carbon in total and a fifth in a single plot.

What the satellite counts. On the HCSA plots E4 found a median of 20 trees per hectare: 2.7% of the stems, 11% of those of 15 cm or more, with a rank correlation of 0.18 against the field count. The 0.05 ha HCSA footprint holds few crowns: 60 of the 148 plots received no tree and 42 received one. The Meta map is a model output stored in whole metres and is smoother than a crown. Without smoothing, every stair step of the integer surface becomes a top and the count rises to 270 per hectare. A tree count from this map is therefore a processing choice, not a measurement, and no setting can be checked without stem lists.

The carbon E4 finds. Table IX reports amounts, not errors. E4 found 25% of the field carbon overall, and 23–26% in each dataset; it under-counts trees and under-sizes them in the same proportion everywhere. Its log correlation with the field value was 0.83 on GEO-TREES and 0.80 on Labrière, so it ranks large plots well. Scaled by the factor learned on other regions (median $\times$3.8), it found 97% of the total, but only 36% of plots fell within $\pm$25% of the field value, against 60% for the map, and on HCSA 9% against 22%. Counting trees adds an interpretable middle step, but it is a worse estimator than learning biomass from the pixels directly.

Why greenness is not enough. Of the 136 plots with a clear Sentinel-2 view in the matching window, the median NDVI (Eq. 7) was 0.77 below 100 Mg ha^{-1}, 0.79 at 100–200, and 0.83 at 200–300 and above 300. Over all plots its rank correlation with AGB was 0.26, and above 150 Mg ha^{-1} it was -0.13. A closed canopy is equally green whatever the biomass beneath it, which is why the models lean on radar, canopy height, and GEDI.

F. Ten Simulated Projects

The tables above score plots. A buyer of credits sees a project. We therefore simulate 1,000 projects of 1,000 ha. Each simulation draws one of the 16 regions with at least 10 plots, each region equally likely, and 10 of its plots at random; the 10 plots stand for the project. For those plots it reports the carbon of the whole project by the field (right grid), the map (E2 with canopy height and GEDI), E4 as is, E4 scaled, and conformal crediting on the map at $\delta = 0.1$ (Section IV-I), as tonnes of CO$_2$e and as a percentage of the field amount. Table X lists the first ten simulations and Appendix A every one of their plots.

Across all 1,000 simulations the map found a median 95% of the field carbon, within $\pm$10% in half of the projects and within $\pm$25% in 80%. E4 never came within 25%. Conformal

TABLE X
TEN SIMULATED 1,000 HA PROJECTS OF 10 RANDOM PLOTS: CARBON FOUND BY EACH GRID, % OF THE FIELD AMOUNT

#	Region	Field, kt	Map	E4	E4 sc.	Credit
1	HCSA Riau	238	71	21	80	64
2	Labrière Nouragues	686	99	23	88	90
3	Labrière Mondah	338	84	16	60	77
4	Labrière Mabounie	594	87	19	71	79
5	HCSA West Papua	254	91	46	180	83
6	HCSA Riau	191	96	8	32	87
7	Labrière Rabi	472	107	33	126	97
8	Labrière Mabounie	556	92	19	72	83
9	HCSA Jambi	287	93	23	88	86
10	Labrière Mondah	156	112	12	45	102
1,000 projects: median			95	23	85	86
5th–95th percentile			46–115	4–44	17–168	42–106
within ±10% of the field			52	0	17	33
within ±25% of the field			80	0	40	76
above the field			38	0	37	12

Field: kt CO_2e from the measured trees, Eq. 1, $\times 0.47 \times \frac{44}{12}$. Map: E2 + canopy height + GEDI. E4 sc.: scaled on other regions. Credit: conformal crediting on the map, $\delta = 0.1$. Seeds 1–10; none drew a GEO-TREES region.

crediting issued a median 86% of the field amount and exceeded it in 12% of projects, near its 10% target; these random 10-plot projects are not the 31 calibration projects of Section IV-I, so the guarantee holds only approximately. The raw map exceeded the field amount in 38% of projects, which is the over-crediting a platform without a bound would issue. Inside a project the plots disagree far more than the totals: in simulation 1 the map gave one Riau plot six times its field value and another half, and the project still came out at 71%.

G. Clouds and Confidence (RQ3)

How often is a plot clear? Table XI gives the plot-level climatology. On a median Indonesian HCSA location, only 21.5% of Sentinel-2 acquisitions see the plot completely clear (Sabah 26%, Gabon and Congo 17%, the Americas 39%). The year is not flat (Fig. 4): in Indonesia the clear share falls to 7.6% in January and 14–15% in December and February, and peaks at 33.5% in May, a factor of 4.4. Revisit compensates for cloudiness. With two Sentinel-2 satellites, the median gap between clear looks is 10 days and the 90th percentile 40 days, so 81% of random dates have a clear look within ±15 days and 96% within ±30. Clouds cluster in time: the measured probability falls below the independence model in every region, most in Africa (±30 days: 79% against 89%).

The scene filter is the real problem. Scene-level cloud percentage agrees only loosely with the plot (Spearman −0.67). In Indonesia, the rule `CLOUDY_PIXEL_PERCENTAGE` < 30 keeps 11.5% of acquisitions, but 34% of the scenes it keeps are cloudy over the plot, and it throws away 65% of the acquisitions in which the plot is clear. At < 40, the threshold our own platform uses for segmentation, 40% of the kept scenes are cloudy over the plot. Masking at the plot instead of filtering scenes gives 2.8 times as many usable clear looks (21.5% against 7.6% of acquisitions) and removes the cloudy ones that pass the filter.

TABLE XI
PLOT-LEVEL SENTINEL-2 CLEAR-SKY CLIMATOLOGY, 2021–2023 (MEDIAN OVER LOCATIONS)

			Gap (d)		P(clear in ±W)	
Region	Loc.	p	Med.	P90	±15 d	±30 d
Indonesia (HCSA)	24	0.215	10	40	0.81	0.96
Sabah (SAFE)	10	0.262	15	41	0.84	0.99
Africa	22	0.166	18	62	0.61	0.79
Americas	40	0.394	10	25	0.82	0.94

p: share of acquisitions with every plot pixel clear. Independence model at ±30 d: Indonesia 0.98, Sabah 0.98, Africa 0.89, Americas 1.00.

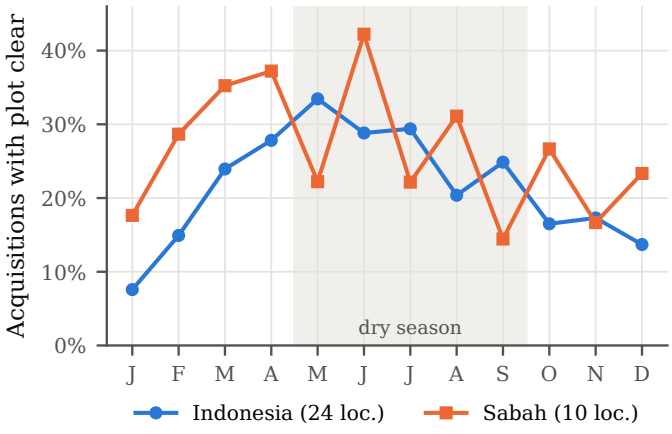


Fig. 4. Share of Sentinel-2 acquisitions in which the whole plot is clear, by month, 2021–2023.

What clouds cost the estimate. Table XII removes the optical block from a growing share of target plots. Because the model leans on radar, terrain, and the annual product, losing every optical feature raises the blind error from 48.0% to 50.0% rRMSE. Modality dropout and the cloud-aware switch both handle partial cloud well; the switch is best on the cloud-hit plots at low q (37.7% against 43.3% for the naive model at $q = 0$, on the 23 plots that really had no clear scene). Imputing optical features from radar is harmful: the bias grows to −61% at full cloud, a warning for SAR-to-optical gap filling used as a model input.

Which intervals are honest? Table XIII compares the four interval methods. Split conformal, the method of the confirmatory run, reaches 88% coverage but is wide (upper bound 16 times the lower) and fails on HCSA (68%). Normalized conformal is narrow but under-covers (79%), because a difficulty model fitted on other regions does not transfer. CQR is the only method that reaches the target overall (89.9%), with intervals three times narrower than split (ratio 5.3) and the best Winkler score. Mondrian calibration by optical availability adds nothing: only 23 plots lack optical data, too few to calibrate a group. HCSA remains under-covered by every method (68–81%): its plots are noisier than any training block, a shift that marginal conformal guarantees do not cover.

Intervals under cloud. Fig. 5 shows the effect. When

TABLE XII
BLIND RRMSE (%) WHEN CLOUDS REMOVE THE OPTICAL BLOCK FROM A SHARE q OF TARGET PLOTS

q	Naive	Dropout	Switch	Impute
0	48.0	48.0	**47.7**	51.5
0.25	48.5	48.6	**47.9**	61.2
0.5	49.6	49.1	**49.1**	67.9
0.75	49.3	49.4	**49.1**	77.5
1	**50.0**	**50.0**	50.4	84.6

TABLE XIII
90% INTERVAL METHODS, LEAVE-ONE-REGION-OUT (COVERAGE; MEDIAN UPPER/LOWER RATIO; WINKLER SCORE)

Method	All	GEO-T.	HCSA	Labr.	Ratio	Winkler
Split	0.884	0.958	0.682	0.989	16.3	4.21
Mondrian	0.882	0.958	0.676	0.989	16.3	4.42
Normalized	0.794	0.782	0.723	0.859	3.3	3.53
CQR	**0.899**	0.937	**0.811**	0.941	5.3	**3.05**

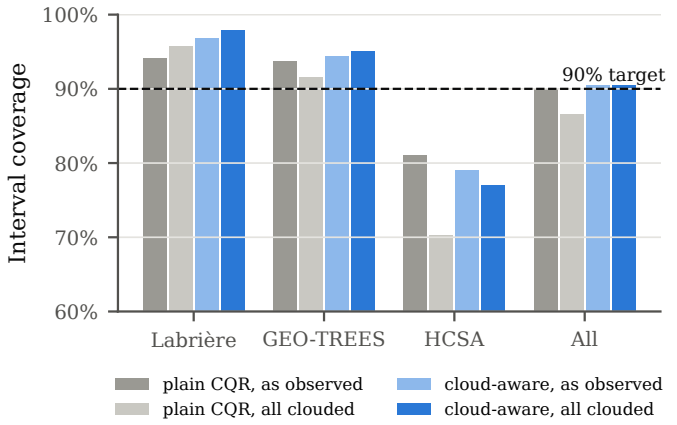


Fig. 5. Coverage of 90% intervals on target plots as observed and with every optical feature removed. Cloud-aware calibration keeps the target overall; plain CQR falls to 86.5%.

every target plot is clouded, plain CQR, calibrated on plots as observed, falls to 86.5% coverage (HCSA 70%). The cloud-aware CQR, calibrated on synthetic cloud, keeps 90.5% (HCSA 77%, GEO-TREES 95%, Labrière 98%). It does this by widening the interval honestly, to a ratio of 7.8 against 5.4. On plots as observed, it covers 90.5% at ratio 5.7. A cloudy plot gets a wider interval, which is what an honest confidence statement should do.

A metadata score is not a confidence. Over all plots, the rebuilt platform-style confidence score correlates with the actual error in the *wrong* direction (Spearman $+0.39$): plots scored as most reliable were the least accurate. That figure is confounded. The recent, cloud-checked HCSA plots score high and are the hardest to predict. Once predicted AGB and the dataset (or the region) are removed, the partial rank correlation is -0.03 to $+0.07$, so the score carries no information about error beyond which dataset a plot belongs to. The learned difficulty $\hat{\sigma}(x)$ fares no better at plot level (partial correlation $+0.02$ to $+0.07$). Neither a metadata score nor a learned difficulty can rank individual plots by error here. Confidence therefore has to be stated as a calibrated interval and checked against field data, as found for global biomass maps [14], [91].

Project-level confidence and over-crediting. For 19 projects (blocks with at least 8 plots), the blind satellite mean missed the field mean by a median of 23%. The naive interval, with plot errors assumed independent, had a median half-width of 28% but covered only 74% of projects (HCSA 33%), and two projects would have been over-credited even after the VM0047 deduction. The two-component interval of Eq. 14 covered all 19, at an honest half-width of 71–86%, which under VM0047 means 61–76% of the credits are deducted. This agrees with earlier coverage studies of model-based estimators that omit residual spatial correlation or model bias [54], [55]; what our test adds is the size of the effect when a model is transferred across regions. A satellite-only project estimate is therefore either over-confident or almost uncreditable.

Field plots change this (Fig. 6, Table XIV). With the satellite as auxiliary data, the ratio estimator is 1.4 to 2.3 times as efficient as the field-only mean on GEO-TREES ($k = 3$ to 20) and about 2 times on Labrière ($k = 3$ to 10); at $k = 20$ the small Labrière regions are nearly fully sampled and the advantage disappears. Equivalently, the satellite replaces roughly half of the field plots needed for a given precision: summed over regions, the plots needed for $\pm 10\%$ at 90% fall from 408 to 216 (GEO-TREES) and from 381 to 162 (Labrière). On HCSA, where the satellite cannot rank small plots (within-region correlation 0.2), the ratio estimator is less efficient than field-only at every k (0.54–0.96), only GREG gains at $k = 20$ (1.25), and the plots needed rise from 1,483 to 1,877. The gain grows with plot size, as found for airborne lidar and InSAR in Tanzania [53]. GREG under-states its own variance at small k (coverage 66–73% at $k = 3$), so the ratio estimator is the safer choice. Finally, the VM0047 deduction does not make an estimate conservative: with $k = 10$ plots, the deducted estimate still exceeded the true stock in 32–43% of draws on the large-plot datasets, for all three estimators, because the deduction only starts above 10% uncertainty. Analytically, the deducted estimate is $\hat{C}(1 - U + 0.10)$, ten points of $\hat{C}$ above the one-sided 95% lower bound, so its exceedance probability must lie between about 5% and 50%; the simulation places it for realistic designs. Critiques of registry deductions so far have been qualitative [86], [92], and over-crediting has been documented for other rules [93].

H. Sequestration (RQ4)

A right-grid failure first. After the reveal, the 27 oil-palm plots showed a mean field stock of 505 Mg ha^{-1}, more than old-growth forest. Every stem in those plots is an oil palm whose recorded "diameter" (median 76 cm) includes the frond bases, and Chave Eq. 4 is a model for dicotyledonous trees. The standard formula is therefore not valid there. The evaluation script had fixed an "excluding oil palm" subset

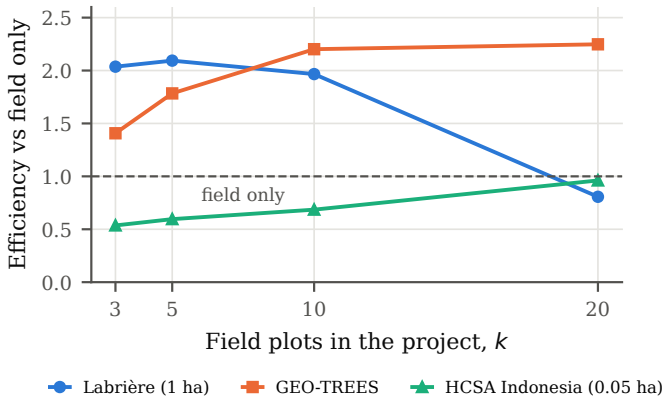


Fig. 6. Efficiency of the ratio estimator (satellite + k field plots) relative to the field-only mean, by empirical MSE over 500 draws. Above 1, the satellite saves field plots.

TABLE XIV
SATELLITE AS AUXILIARY DATA: COVERAGE OF 90% CI AND EFFICIENCY VS FIELD-ONLY (500 DRAWS)

		Field only		Ratio (satellite + k)		
Dataset	k	**Cov.**	**RMSE**	**Cov.**	**RMSE**	**Eff.**
GEO-TREES	5	0.91	16.3	0.90	12.2	1.78
	10	0.88	10.4	0.89	7.0	2.20
	20	0.90	5.7	0.90	3.8	2.25
HCSA	5	0.85	36.7	0.80	47.6	0.60
	10	0.86	22.4	0.86	27.1	0.69
	20	0.89	11.1	0.87	11.3	0.96
Labrière	5	0.89	17.9	0.86	12.3	2.09
	10	0.88	10.1	0.88	7.2	1.97
	20	0.88	3.1	0.89	3.5	0.81

RMSE in % of the project mean. Eff.: MSE(field)/MSE(ratio); >1 saves plots.

before the reveal, and we report it as the primary result: 196 plots in 15 blocks, mean field ΔAGB $= 1.87\,\mathrm{Mg\,ha^{-1}\,yr^{-1}}$ ($0.88\,\mathrm{t\,C\,ha^{-1}\,yr^{-1}}$), median census gap 8.4 years.

Blind satellite change estimates do not beat "no change". Table XV gives the scores. At plot level the Chloris difference (S0) has almost the same error as predicting zero (RMSE 8.06 vs $8.12\,\mathrm{Mg\,ha^{-1}\,yr^{-1}}$, $r = 0.07$). The E2 stock difference (S2) is twice as bad and has the wrong sign: it predicts a loss of $4\,\mathrm{Mg\,ha^{-1}\,yr^{-1}}$ where the forest gained 1.9. The cause is a sensor artefact: Sentinel-1 existed for only 13% of the plots at t_0 but for all of them at t_1, and no L-band mosaic before 2015 is available in the catalogue, so the two stock predictions are made from different inputs. Restricting the stock model to sensors present at both dates (S2c: Landsat and terrain) halves the bias but not the noise.

Why. The blind E2 stock error on these plots has a standard deviation of $146\,\mathrm{Mg\,ha^{-1}}$. If the errors at t_0 and t_1 were independent, differencing them over 8.4 years would add noise of about $25\,\mathrm{Mg\,ha^{-1}\,yr^{-1}}$, thirteen times the signal. The errors are partly correlated, because the same place is seen twice, which is why the observed S2c error ($14\,\mathrm{Mg\,ha^{-1}\,yr^{-1}}$) is lower, but it is still seven times the signal. A stock estimator

TABLE XV
BLIND SEQUESTRATION, SAFE EXCLUDING OIL PALM (MG AGB $\mathrm{HA^{-1}\ YR^{-1}}$; FIELD MEAN 1.87)

	Plot (196)			**Block mean** (15)	
Estimator	**RMSE**	**Bias**	r	**MAE**	**Bias**
No change	8.12	−1.87	–	2.05	−1.92
S0 Chloris difference	8.06	−1.01	0.07	1.61	−1.20
S2 E2 stock difference	16.36	−5.90	−0.02	7.40	−7.08
S2c consistent sensors†	13.96	−3.50	0.09	4.48	−3.77
S3 change model, LOBO†	8.39	−0.50	−0.07	1.33	−0.27
Regional default rate†	**7.91**	**0.00**	–	**0.98**	−0.05

†Exploratory, defined after the reveal.

with plot-scale error of 30–100% cannot resolve growth of 1% per year.

What does work is not a satellite. Two estimators trained on the other SAFE blocks' measured change (leave-one-block-out, run after the reveal and therefore exploratory) were added: a LightGBM on Landsat and Chloris change features (S3), and the plain mean rate of the other blocks with no satellite input at all. The regional default rate is the most accurate estimator at both scales (block-mean MAE 0.98 vs 2.05 for no change), and S3 learns little more than that rate. Such a rate is in effect a local Tier 1 removal factor of the kind compiled by Requena Suarez *et al.* [94]. At this plot size and with these sensors, the information about growth comes from nearby field remeasurements, not from the pixels. Adding the Earth Engine inputs did not change this. These were the 2010 PALSAR mosaic, which makes the L-band record consistent at both censuses, CCI biomass change, and Hansen forest loss between the censuses. A change model on them lost to the regional default rate (block MAE 1.20 vs 0.98), and Hansen loss separated plots that lost biomass only weakly (AUC 0.57).

Out-of-region stock on Borneo. The same E2 model, trained without any Bornean plot, misses SAFE stock at t_0 by 132% rRMSE with a +80% bias (excluding oil palm). It over-predicts the 0.06 ha logged and riparian plots (field mean 99, predicted $217\,\mathrm{Mg\,ha^{-1}}$), a second instance of the small-plot problem seen on HCSA.

I. From Intervals to Credits (RQ5)

Table XVI credits the 31 projects under each rule at $\delta = 0.10$.

Satellite only. Crediting the raw map over-credits 35% of projects. The VM0047 deduction applied to the honest two-component uncertainty never over-credits, but it credits only 48% of the true stock. Conformal crediting meets its target: 9.7% of projects are over-credited (exact 95% interval 2–26% with 31 projects), and 72% of the true stock is credited without a single field plot in the project.

With field plots. The VM0047 deduction over-credits 23% of projects with 5 plots and 29% with 10, up to 42% of the large-plot projects. The design lower bound at the same δ realises 8.5–9.9% and gives up only 1.5–4 points of yield. Plain conformal calibration meets the target on average, but

TABLE XVI
CREDITING RULES AT TARGET $\delta = 0.10$: REALISED OVER-CREDITING AND YIELD (31 PROJECTS, CROSS-REGION CALIBRATION)

	Over-credited			
Rule	**All**	**Large**	**Small**	**Yield**
Satellite only (no plot in the project)				
No deduction	35.5%	39.1	25.0	94.4%
VM0047, two-component U	0.0%	0.0	0.0	47.8%
Parametric, two-component	0.0%	0.0	0.0	59.5%
Parametric, independent	9.7%	–	–	72.6%
Conformal (Eq. 16)	9.7%	13.0	0.0	72.2%
Satellite + 5 random field plots (ratio estimator)				
VM0047 ($U - 10\%$)	22.6%	27.6	8.1	82.2%
Design lower bound	9.9%	10.2	9.2	80.7%
Conformal, plain	9.4%	3.8	25.3	80.8%
Conformal, studentized	9.6%	9.2	10.6	80.7%
Satellite + 10 random field plots (16 projects)				
VM0047 ($U - 10\%$)	29.3%	42.1	12.9	90.0%
Design lower bound	8.5%	9.4	7.4	86.1%
Conformal, studentized	6.5%	6.1	7.0	82.1%

Large: 1 ha and network plots (23 projects); small: 0.05 ha HCSA plots (8). With field plots: mean over 200 draws.

TABLE XVII
SATELLITE-ONLY BOUNDS APPLIED TO FIVE MAPS ($\delta = 0.10$): REALISED OVER-CREDITING / YIELD

Map	**Param., indep.**	**Param., two-comp.**	**Conformal**
E2 (blind LightGBM)	9.7 / 72.6	0.0 / 59.5	9.7 / 72.2
Chloris product	19.4 / 77.6	3.2 / 53.7	12.9 / 66.2
E2–Chloris average	6.5 / 75.5	0.0 / 60.8	9.7 / 79.3
E2 × 1.1	16.1 / 79.8	3.2 / 65.4	9.7 / 72.2
E2 × 1.2	19.4 / 87.1	6.5 / 71.3	9.7 / 72.2
Worst excess over δ	+9.4	−3.5	+2.9

Percent of projects over-credited / percent of true stock credited.

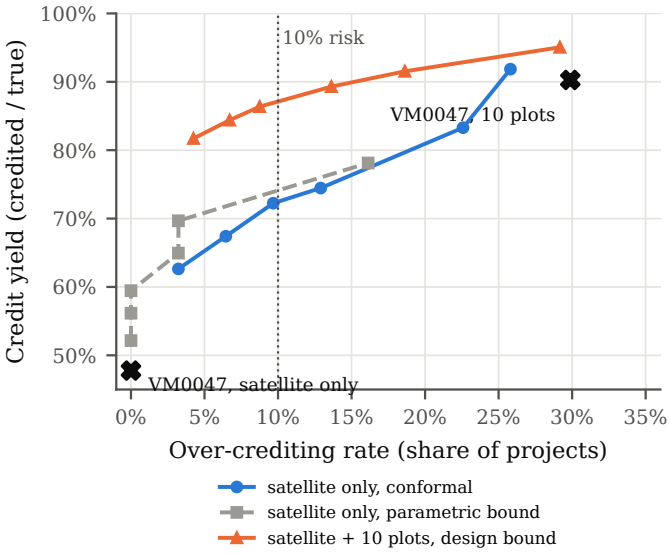


Fig. 7. Integrity-yield frontier: realised over-crediting against credit yield as the target δ varies from 0.05 to 0.30. Crosses mark the VM0047 deduction.

not by plot size: with 5 plots, it over-credits 25% of small-plot projects and 4% of large-plot ones. Studentization repairs this (10.6% and 9.2%), matching the design bound, whose validity comes from random sampling. Once plots are drawn at random, the classical design bound is the efficient choice, and conformal calibration adds a check, not credits.

Is the bound calibrated whatever map it is given? The satellite-only result above could be luck, because E2 under-predicts by about 9% on average and so rarely over-credits. Table XVII applies the same rules to five maps: E2, the Chloris product, their average, and E2 inflated by 10% and 20%, as an optimistic vendor map might be. Over the five maps, the conformal bound's realised over-crediting stays within 2.9 points of the target at $\delta = 0.10$, and within 2.6 points at $\delta = 0.20$. The independent parametric bound overshoots by up to 9.4 and 15.5 points. The two-component bound is conservative on E2 and drifts upward as the map is inflated. Inflation also shows the incentive. Under the parametric bound, a map inflated by 20% earns 87% of the true stock instead of 73%, and doubles the over-crediting rate. Under conformal crediting it earns exactly the same 72.2%, as the invariance property predicts. With the sharper map of Table VI (canopy height and GEDI), conformal crediting met the same target (9.7%) and credited 84% of the true stock instead of 72%. A better map buys credits; it does not buy risk.

Integrity-yield frontier. Fig. 7 traces each rule as δ varies. Both VM0047 points lie below the frontier. With 10 plots, VM0047 takes 29% risk for 90% yield, where the design bound gives 92% at 19%. Satellite-only, it credits 48% at no realised risk, where the other bounds give 60–72% at 0–10%. The frontier also prices a field plot. At about 10% realised risk, the satellite alone earns 72% of the true stock, 5 random plots raise it to 81%, and 10 plots to 86%. The parametric two-component bound lies slightly above the conformal curve at low risk. It is not better calibrated: its realised risk sits far from its nominal one, in a direction that changes with the map (Table XVII). A user of the parametric bound cannot know beforehand what risk is being taken.

What the guarantee needs. The guarantee is marginal over the population of projects in the calibration bank, and it is only as good as that bank. With a mixed bank, network-plot projects were over-credited in 33% of cases at $\delta = 0.10$, against 0% for the 1 ha and HCSA projects. Calibrating only on projects from other datasets raised this to 44%. Calibrating only on the same dataset brought it to 0–14%, but with 6–8 calibration projects the quantile reached the largest score, and the mean yield fell to 18%. A conditional guarantee for a project type therefore needs at least 9 field-verified projects of that type for $\delta = 0.10$, and 19 for $\delta = 0.05$. Two further limits apply. The truth in the bank is itself a field estimate, so the guarantee is relative to that reference. And projects with field truth may not be selected like the projects being credited, which is the main threat to exchangeability.

Position. The conservativeness target is that of Grassi *et al.* and Neeff [86], [87], and the machinery is split and nor-

TABLE XVIII
RELIABILITY: REGION-BLOCK BOOTSTRAP 95% INTERVALS (2,000 DRAWS) AND SEED SPREAD

Quantity	Value	95% interval
rRMSE E2, all plots	48.5%	34.9–68.7
rRMSE E2, Labrière (1 ha)	28.6%	22.8–38.4
rRMSE E2, GEO-TREES	41.9%	35.9–51.8
rRMSE E2, HCSA	102.9%	88.1–121.7
E2 − Chloris (points)	−0.8	−8.3 to +5.3
E2–Chloris average − E2 (points)	−3.7	−7.2 to −0.6
Radar only − optical only (points)	−2.3	−11.1 to +5.1
CQR coverage, as observed	89.9%	83.9–94.4
Cloud-aware CQR coverage, fully clouded	90.5%	84.7–95.2
Cloud-aware − plain, fully clouded (points)	+4.0	+1.9 to +6.6
E2 rRMSE over 5 seeds (mean ± SD)	48.8	±0.36

malized conformal prediction [43], [44]. Conformal intervals for area-level parameters have been proposed for small-area estimation [95], and for the true effect of a new trial in meta-analysis [96]. What we have not found before is the following: a crediting rule calibrated across projects against field truth, its empirical test under spatial transfer and biased maps, its invariance to map inflation, and a measured integrity-yield trade-off for the registry deduction.

J. How Reliable Are These Numbers?

Every score above comes from 27 region blocks, and errors are correlated within a block. We therefore resampled whole blocks with replacement (2,000 draws) and refitted E2 with five LightGBM seeds (Table XVIII). Seed noise is small: the overall rRMSE moved by 0.36 points (SD). Region sampling is not: the 95% interval of the overall rRMSE runs from 35% to 69%. Three kinds of claim survive this test. The first is the differences between datasets and plot sizes, whose intervals do not overlap. The second is the gain of the E2–Chloris average over E2 (-3.7 points, interval -7.2 to -0.6). The third is the coverage gain of cloud-aware calibration under full cloud ($+4.0$ points, $+1.9$ to $+6.6$). Two claims do not survive. The difference between E2 and Chloris (-0.8 points, -8.3 to $+5.3$) and between radar and optical inputs (-2.3, -11.1 to $+5.1$) are within noise. We report them as tendencies, not findings.

K. Systems (RQ6)

Production runs. The stock extraction ran 2,052 tasks on 24 workers in 2.1 h (0.26 tasks s^{-1}) with a measured load factor of 1.47, close to the 1.40 predicted from task counts; 85 transient errors were absorbed by retries. We stopped the first attempt, which used tile-only placement, to switch policies; the restart skipped the 92 tasks already in the content-addressed cache. Seventeen tasks (0.8%) failed permanently: 13 hit two Landsat scenes whose files could not be read from the catalogue, and four hit unreadable Sentinel-2 files. Retrying cannot repair a permanently broken input; the runner records such tasks and the model treats their features as missing, and a scene-level fallback is the obvious next step. The SAFE extraction ran 2,006 tasks on 16 workers in 2.3 h with no permanent failures, 36 retries, and a load factor of 1.44.

TABLE XIX
STRONG SCALING AND FAULT INJECTION (48 TASKS, EMPTY CACHE)

Workers	Fault p	Wall (s)	Speed-up	λ	Identical
1	0	498.5	1.00	1.00	47/47
2	0	207.8	2.40	1.00	47/47
4	0	144.8	3.44	1.00	47/47
8	0	98.4	5.06	1.00	47/47
16	0	93.3	5.34	1.33	47/47
32	0	91.4	5.45	2.67	47/47
8	0.1	106.7	4.67	1.00	47/47
8	0.2	105.0	4.75	1.00	47/47
8	0.3	105.7	4.72	1.00	47/47

Scaling, faults, and resume. A random sample of 48 tasks (Landsat, Sentinel-1, PALSAR, terrain) was run on 1 to 32 workers, each time on an empty cache (Table XIX). Throughput rose 5.1-fold up to 8 workers and then flattened (5.5-fold at 32). Two limits explain this. With 48 tasks, 32 workers hold one or two buckets each, so the load factor rose to 2.67. And the makespan cannot fall below the slowest single task, a multi-scene read, however many workers wait beside it. The extraction is bound by remote reads, not by local compute. Every run was compared with the single-worker run task by task: all 47 finished results were bit-identical at every worker count. With transient faults injected at rates 0.1, 0.2, and 0.3 per attempt, the runner absorbed 13, 18, and 28 retries, and again all 47 results matched. The one task that never finished, in any run, is the one whose Landsat file is unreadable, which is a permanent fault. In the resume test, a run stopped after 23 tasks; the restart skipped those 23 from the cache, ran the remaining 24, and produced output identical to the uninterrupted run.

The super-linear step from one to two workers reflects variable latency of the remote service between runs; each configuration was run once, so differences of a few seconds are within noise.

Ledger. The final log holds 37 entries : four seals (one per right-grid build and one per evaluation run), three commits, two reveals, and 28 label releases, none of which included a label from its target block. The chain verifies end to end. In unit tests, editing any entry broke verification, a commit after a reveal was refused, and a reveal with a modified outcome file was refused.

V. DISCUSSION

A. Can Satellites Replace the Field Plot?

Not at the scale of a plot, and not for growth. Tested blind, open satellite data put a 1 ha plot's carbon stock within roughly 20–30% of the field-formula value once canopy height and GEDI are included, and a 0.05 ha plot within about 100%; the best honest interval (CQR) is a factor of five wide. For sequestration, the quantity on which credits are paid, no satellite estimator we built beat assuming no change, and a regional average of measured growth beat all of them. A platform that issues plot-level or annual growth numbers from

TABLE XX
BLIND RESULTS OF THIS STUDY AGAINST PUBLISHED BENCHMARKS

Study	Sensor	Reference	Error
This study, Labrière (blind)	multi-sensor	1 ha plots	28.6%
This study, + canopy ht. + GEDI	+ 1 m CHM, GEDI	1 ha plots	20.6%
This study, GEO-TREES (blind)	same	network	41.9%
This study, HCSA (blind)	same	0.05 ha plots	102.9%
Berninger 2018, Kalimantan [97]	S1 + PALSAR	lidar layer	31–38%
Jucker 2018, Sabah [98]	airborne lidar	0.1 / 1 ha	25 / 13%
Araza 2022, CCI [14]	CCI map	0.1° cells	29–61*
Santoro 2021 [4]	GlobBiomass	110 k plots	SD ≈50%

*RMSD in Mg ha^{-1} for AGB below 300; 107 at 300–400. Other errors are rRMSE.

these sensors alone is reporting a default constant, whether or not it says so.

Nor can a satellite count trees. The field ground truth is a list of stems with diameters, heights, and species. The 1 m canopy height map shows the top of the canopy: it found 3% of the stems, cannot see the 75% of stems below 15 cm or anything under the crowns, and cannot see species, which alone moves a plot's carbon by a fifth. Applying the field formula to what the map shows finds a quarter of the carbon. The satellite's real "fixed constant" is canopy height, not a tree count, and the step from height to carbon has to be learned from field plots.

What satellites do well is replace part of the inventory. Used as auxiliary data in a ratio estimator, they were 1.4–2.3 times as efficient as field plots alone on large-plot datasets, roughly halving the plots needed for a given precision (Table XIV). The useful framing is not "satellites instead of field plots" but "satellites plus a small, well-designed set of field plots instead of a full inventory". Even that only holds where the satellite can rank plots within a project. On the small Indonesian plots it cannot (within-region correlation 0.2), and there it does not save a single plot.

B. Our Numbers Against the Literature

Table XX places our blind results next to published validations. Our 1 ha error (29%) is close to the 31–38% reported for Sentinel-1 and PALSAR in Kalimantan [97]. That study was validated against a lidar-upscaled reference layer, not blind against field plots, and it reports the same underestimation above 200 Mg ha^{-1}. Our saturation near 300 Mg ha^{-1} matches the flattening near 250 Mg ha^{-1} in the global map of Santoro *et al.* [4] and the steep rise of map error above 300 Mg ha^{-1} found by Araza *et al.* [14]. Airborne lidar in Sabah, the best remote sensing can do, still errs by 25% at 0.1 ha and 13% at 1 ha [98]. Plot size dominates every comparison. Local variation of AGB alone has a coefficient of variation of 46% at 0.1 ha and 17% at 1 ha [13], and Duncanson *et al.* recommend plots of at least 1 ha for validation [51]. If we remove that expected representativeness noise in quadrature, the blind satellite error is about 23% on the 1 ha units and still about 90% on HCSA. Plot size alone does not explain the Indonesian result. The mosaic of open land, scrub, and forest around HCSA plots, and a GPS error comparable to the 12.6 m plot radius [51], are the likely remainder.

C. Clouds: What Actually Matters

Three findings change how clouds should be handled. First, *how often* a plot is clear matters less than *how* clearness is decided. A clear look within ±30 days exists for 96% of dates in Indonesia, because revisit compensates for an 80% cloud frequency [32], [34]. The scene-level filter that most pipelines apply, by contrast, discards two thirds of the clear looks and keeps a third of cloudy ones. Pixel-level masking at the plot, with SCL, Fmask, or Cloud Score+ [35]–[37], nearly triples the usable data. Second, *season matters*: in Indonesia the clear share in May is 4.4 times that in January, so field campaigns and their satellite windows should be planned for the dry months. Third, a model led by radar and an annual product loses only two points of accuracy without any optical data, so clouds are an availability problem for optical indices but not for a well-designed multi-sensor stock estimate. That is the practical form of the SAR–optical fusion argued for by Reiche *et al.* [41]. Filling optical gaps from radar and feeding them to the model, however, was the worst option we tested. Reconstructed reflectance is a model output whose error does not disappear because the image looks complete.

D. Confidence That Means Something

Honest confidence has three layers, and each failed in a characteristic way. At *plot* level, conformal intervals were honest overall, but only CQR was both honest and usable, and missing-data-augmented calibration [8] kept it honest when optical data were missing, outside the assumptions under which that method is guaranteed. The guarantee is marginal: under the shift from large to small plots (HCSA), coverage fell to 68–81% for every method. Spatially weighted conformal prediction [46], [47] is the next step. At *project* level, independence is the dangerous assumption. Map errors are correlated over tens of kilometres [14], a project shares one regional bias, and an interval that ignores it covered only 74% of projects. The two-component interval is honest, and it shows that a satellite-only project estimate would lose most of its credits under VM0047. At the level of the *standard*, a deduction of $U - 10\%$ is not a lower confidence bound. With ten field plots, the deducted estimate still exceeded the true stock in a third of draws. Conformal crediting (Section IV-I) makes the over-crediting rate a design parameter rather than an accident, and removes the reward for inflating the map.

A metadata quality score answers a different question from a confidence interval. Ours rewarded recent, cloud-checked imagery. Over all plots it trusted most the plots the model got most wrong, and within a dataset it said nothing at all about error. Confidence has to be learned from errors against field data and reported in the units of the claim, as an interval in t C ha^{-1}, not as a grade.

E. The Ground Truth Is a Model Too

The right grid is fixed by the standards, but the standards leave real choices, and those choices are as large as the precision targets they enforce. Switching the root-to-shoot ratio from the SNI 7724 value (IPCC 2006) to the 2019 Refinement

changes total carbon by 13%. Switching from the pantropical Chave equation to the East-Kalimantan equation of Basuki *et al.* leaves the national total unchanged but moves provincial stocks from -40% to $+121\%$. AR-TOOL14 counts sampling error only, yet measurement and allometry carried up to 64% of the variance of a provincial mean. The same data receive deductions from 0 to 34% depending on which standard is applied. For growth the field answer itself is fragile. With 9–16 plots of 0.06 ha, VM0047 Eq. 28 deducts most or all of the credit unless the correlation of permanent plots is used, and the plot-level measurement error of the change halves only when the same stem is assessed with the same equation at both dates. The target "100% accuracy" therefore has no meaning. The meaningful target is to fall within the uncertainty of a field estimate made to a stated standard, and that uncertainty must itself be computed in full.

F. The Path Toward the Field Answer

The experiments rank the levers. (i) *Report at project scale*: aggregation removes most random error. (ii) *Add local plots and use the satellite as auxiliary data*, which is the most effective step, provided the plots are large enough to be ranked by the satellite. (iii) *Mask clouds at the plot, not the scene*, and match the window to the season. (iv) *Lead with structure-sensitive sensors*. Radar was ahead of optical on both large-plot datasets, though within sampling noise, and it is what keeps the estimate alive under cloud; adding a 1 m canopy height map and GEDI cut the blind error by 9 points (Table VI), and drone lidar (Section V-G) and the 2025 L- and P-band missions NISAR and ESA BIOMASS are the next steps. (v) *Keep sensors consistent through time*: half of the sequestration error came from Sentinel-1 appearing between the two censuses. (vi) *Combine rather than choose*: averaging the transfer model with an independent global product was the best blind stock estimate. (vii) *Calibrate confidence*: CQR with cloud-aware calibration at plot level, the two-component variance at project level, and credit the lower bound.

G. Toward a Multi-Tier System: Drones

Three results of this study point in one direction. The biggest gain in accuracy came from fine-scale canopy structure: a 1 m canopy height map. Plots smaller than about 0.25 ha cannot be predicted by anything we tried. Trees cannot be counted on a satellite canopy height map (Section IV-E). And the most effective step toward the field answer is more field plots. NusaCarbon's planned drones target all three. We set out the design and what each tier still has to prove; no drone data enter the results above.

Under the canopy. The swarm replaces the slowest step of the right grid: measuring every stem's diameter. Under-canopy laser scanning from a piloted drone has reached a DBH error of 0.6–0.9 cm (2–3%) in boreal plots [99]. Fully autonomous flight reached 3.8% in a simple stand but 20–22% in complex ones [100]. Swarms of micro drones already navigate dense forest autonomously [101], but none carries a measuring payload yet, and we found no under-canopy DBH study in tropical forest. Species, needed for wood density, can fall back on genus means, which carry 74% of species-level variation [103]. Drone imagery has classified key restoration species in Indonesia with 74–92% accuracy [104]. Plugged into Step 5 of the right grid, the swarm's own measured DBH error replaces the Chave 2004 error model, so its uncertainty is propagated rather than assumed.

Above the canopy. The Zenmuse L2 gives a canopy height model at centimetre scale over about 2.5 km^2 per flight, according to the manufacturer. We found no peer-reviewed test under tropical canopy. Airborne lidar in Sabah reached a plot error of 13% at 1 ha, against far larger errors for satellite maps at the same plots [98]. In Peru its uncertainty fell to about 5% at 5 ha [105]. A chain from field plots to drone lidar to Sentinel-2 has beaten direct field-to-satellite models, and it needed about 37% fewer field plots [106]. IPCC accepts biomass maps built from field and remote-sensing data at Tiers 2 and 3 [107]. A repeat flight measures canopy-height change on the same trees, which is the quantity satellites failed to resolve here.

How to test it. The protocol of this paper applies unchanged. Under-canopy plots, computed by the standard chain, are the sealed right grid. Drone-lidar and satellite estimates are the left grid. The ledger, conformal intervals, and conformal crediting carry over. Three hypotheses should be tested on Indonesian 1 ha plots before any claim is made. (H1) Swarm DBH is within 3% of tape on tropical stems, with at least 90% of stems detected. (H2) L2-based AGB is within 15% of the right grid at 1 ha, leave-one-site-out. (H3) Repeat-flight growth beats both zero change and the regional default rate at block level. If H2 holds, the satellite tier can be recalibrated on drone-lidar maps instead of sparse plots, which is how GEDI's own under-represented Southeast-Asian strata could be filled [25].

H. Implications for NusaCarbon

Our audit found default constants where measurements were expected, a scene-level cloud filter, and a confidence grade computed from image metadata. The evidence suggests a concrete redesign:

- Mask clouds per pixel at the plot (SCL or Cloud Score+) instead of `CLOUDY_PIXEL_PERCENTAGE`, and widen the window in the wet season.
- Estimate stock with a radar-led multi-sensor model trained with modality dropout, and attach a cloud-aware CQR interval instead of the letter grade.
- Report projects with the two-component interval, and calibrate each project with ten to twenty large field plots through a ratio estimator.
- Credit with a bound whose risk is calibrated: conformal crediting from a calibration bank of field-verified projects when no plot is measured, the design lower bound when random plots are, never the $U - 10\%$ deduction alone.
- Credit sequestration from field remeasurement of permanent plots, and use satellites to detect loss and disturbance.

TABLE XXI
PLANNED NUSACARBON TIERS: WHAT EACH MEASURES, PUBLISHED EVIDENCE, AND WHAT REMAINS TO BE SHOWN

Tier	Sensor	Role	Published evidence	To be shown
Under canopy	drone swarm with laser scanner (to be built)	Measure D and stem position on 1 ha plots without a field crew; becomes the right grid	Under-canopy UAV laser scanning: DBH RMSE 0.6–0.9 cm (2–3%), stems detected 84–93%, boreal, piloted [99]; autonomous flight 3.8% to 22% by stand complexity [100]; swarms navigate forest [101] but none has published DBH accuracy	DBH accuracy and stem detection under tropical canopy; full autonomy at 1 ha
Above canopy	DJI Matrice 350 RTK + Zenmuse L2 lidar	Canopy height over the whole project, converted to AGB with the under-canopy plots; repeat flights for change	L2: vertical accuracy 4 cm at 150 m, up to 5 returns, about 2.5 km^2 per flight (manufacturer); airborne lidar in Sabah: 13% plot error at 1 ha [98]; Kalimantan lidar–AGB $R^2 \approx 0.8$ [102]	AGB error at 1 ha in Indonesian forest; ground detection under dense canopy; growth from repeat flights
Satellite	Sentinel-1/2, Landsat, PALSAR, GEDI, 1 m canopy height	Wall-to-wall stock, yearly disturbance monitoring, screening; calibrated by the drone tiers	This study: 21% at 1 ha, 7–9% on project totals outside Indonesia, no skill for growth	Gain when calibrated by drone lidar instead of sparse plots

- Log every estimate through the commit-reveal ledger.
- Fix the GEDI band that is read as canopy height.

I. Threats to Validity

Internal.

- The right grid is itself a model: Chave allometry carries its own error, and GEO-TREES and Labrière AGB were computed by their authors, not by us.
- HCSA is released as a 2023 dataset without per-plot dates, and Labrière plots have campaign-year dates only, so temporal matching for those plots is at year precision.
- The standard formula failed on SAFE oil palm. Those plots are excluded by a subset fixed before the reveal, and 45 palm stems remain in forest plots.
- The improvement ladder, the change models, the cloud and interval experiments, the project-level analysis, E4, and the project simulations were defined after the reveals and are exploratory. Only the E0, E2, S0, and S2 scores are confirmatory, and we report every variant we ran; Section IV-J states which differences exceed sampling noise.
- The ledger proves the order of events, not the absence of leakage upstream of it. GEDI biomass models were calibrated on field plots and airborne lidar that may include some of our sites, notably in Gabon; the canopy-height maps were trained on lidar outside our plots, and most of the gain in Table VI comes from them. The Chloris product, used as E0 and as an input of the ensemble, was calibrated by its makers on field and lidar data that we cannot inspect; if any of our plots were among them, that path leaks. E2 without the product is free of this path.
- Provincial stratum weights are plot shares, because mapped stratum areas are not public.
- The cloud climatology uses SCL, whose own overall accuracy is about 80–90% [36].

Construct.

- Plot-scale truth (0.05–1 ha) is compared with 10 m to 4.6 km pixels, and GPS error on small plots adds noise that no estimator can remove.
- Conformal intervals calibrated on plots cover plot values, not true stocks.
- GEDI biomass models were calibrated on field plots elsewhere, so E1 is free of our field data but not of field data altogether.

External. Indonesian plots come from one programme, collected by trained non-foresters; results may not transfer to other forest types or survey teams. Projects in the project-level analysis are region blocks, not registered projects.

VI. CONCLUSION

We asked whether open satellite data can replace the field plot in forest carbon MRV, and tested it blind. A sealed right grid was computed step by step to SNI 7724, IPCC, CDM, Verra, and Gold Standard rules with a full uncertainty budget. A left grid saw only coordinates and dates. A hash-chained ledger makes any change to a prediction after the answer was opened detectable.

Across 476 stock plots, the blind estimate missed by 49% relative RMSE overall with Sentinel, Landsat, and PALSAR, and by 40% after adding a 1 m canopy height map and GEDI lidar: 21% on 1 ha plots, 7–9% on project totals outside Indonesia, and still about 100% on small Indonesian plots. Sharper canopy structure, not a better algorithm, gave the largest gain, which is the case for drone lidar. Counting trees on the 1 m map and applying the field formula found 3% of the stems and a quarter of the carbon, and in 1,000 simulated projects the best map found a median 95% of the field carbon, within 25% in four projects of five. In Indonesia only 21.5% of Sentinel-2 acquisitions see a plot clear. Masking at the plot instead of filtering scenes nearly triples the usable data, and a radar-led model loses only two points without optical data. Missing-data-augmented conformal calibration kept 90% intervals honest under full cloud, while a metadata confidence score carried no information about error beyond

dataset differences. At project scale, an interval that ignores spatially correlated error covered only 74% of projects; with field plots as calibration, the satellite was 1.4–2.3 times as efficient as field plots alone on large-plot datasets. Conformal crediting met a stated over-crediting risk without a single field plot in the project, earning 72% of the true stock at 10% risk, was invariant to map inflation, and showed that the registry deduction sits below the integrity-yield frontier. For annual sequestration, no satellite estimator beat "no change", and even the field answer was too uncertain to credit without the correlation of permanent plots.

Satellites are a stock-mapping, disturbance, and plot-saving tool, not a replacement for the field plot, and their confidence must be calibrated, not graded. The contribution is the evaluation, not a new method: the protocol, the ledger, and the extraction runner are reusable. Together they turn a platform's accuracy claim into something a third party can check.

Appendix A
Per-Plot Rows of the Ten Simulated Projects

Tables XXII and XXIII list every plot of the ten simulations of Table X. Stems: field stems per hectare from the HCSA stem list ("–" where only plot biomass is published). Trees: trees per hectare that E4 counted on the 1 m canopy height map. Carbon columns are $\mathrm{t\,CO_2e\,ha^{-1}}$; Map % is the map as a percentage of the field value. Rows for all 475 plots are in `tree_benchmark_plots.csv` and in the dashboard.

TABLE XXII
PER-PLOT ROWS, SIMULATIONS 1–5

Sim	Region	Plot	Stems	Trees	Field	Map	Map %	E4	E4 sc.
1	HCSA Riau	HCSA_Riau_RU05	960	0	462	237	51	0	0
		HCSA_Riau_RU21	500	0	10	63	611	0	0
		HCSA_Riau_RU06	720	0	170	120	71	0	0
		HCSA_Riau_RU15	580	20	114	463	407	30	117
		HCSA_Riau_RU01	1960	0	833	152	18	0	0
		HCSA_Riau_RU22	60	40	12	78	649	23	90
		HCSA_Riau_RU10	220	40	79	87	111	29	113
		HCSA_Riau_RU19	420	0	45	91	203	0	0
		HCSA_Riau_RU14	120	20	49	204	419	194	745
		HCSA_Riau_RU03	160	20	602	181	30	218	839
2	Labrière Nouragues	LAB_NOU04h5	–	15	719	789	110	168	631
		LAB_NOU02h10	–	23	748	755	101	201	752
		LAB_NOU03h2	–	15	893	627	70	126	474
		LAB_NOU04h9	–	16	876	719	82	175	656
		LAB_NOU02h6	–	22	532	690	130	149	561
		LAB_NOU02h5	–	18	497	637	128	139	522
		LAB_NOU04h12	–	19	805	657	82	183	687
		LAB_NOU02h1	–	19	490	560	114	106	397
		LAB_NOU04h2	–	24	508	678	134	218	818
		LAB_NOU02h9	–	18	795	712	90	142	531
3	Labrière Mondah	LAB_MON11h	–	23	60	59	98	6	22
		LAB_MON05h	–	18	161	167	104	13	50
		LAB_MON22h	–	9	498	453	91	32	122
		LAB_MON01h	–	20	44	49	112	4	15
		LAB_MON21Ah	–	27	276	236	85	54	202
		LAB_MNG03h	–	29	880	496	56	164	617
		LAB_MNG04h	–	21	765	559	73	152	574
		LAB_MON14h	–	23	258	281	109	41	154
		LAB_MON19h	–	3	6	9	145	0	1
		LAB_MON13h	–	17	427	534	125	74	281
4	Labrière Mabounie	LAB_MAB05h	–	9	756	717	95	106	400
		LAB_MAB11h	–	18	902	701	78	166	625
		LAB_MAB03h	–	0	575	187	33	0	0
		LAB_MAB10h	–	4	594	200	34	11	40
		LAB_MAB12h	–	16	296	544	184	121	456
		LAB_MAB04h	–	18	791	659	83	209	788
		LAB_MAB07h	–	12	426	550	129	88	334
		LAB_MAB01h	–	23	565	489	87	134	505
		LAB_MAB08h	–	22	501	576	115	150	566
		LAB_MAB06h	–	14	533	548	103	129	488
5	HCSA West Papua	HCSA_West_Papua_PB12	580	0	89	108	121	0	0
		HCSA_West_Papua_PB01	860	0	192	220	114	0	0
		HCSA_West_Papua_PB19	1260	0	556	125	22	0	0
		HCSA_West_Papua_PB09	600	20	532	301	57	399	1546
		HCSA_West_Papua_PB20	1340	0	224	356	158	0	0
		HCSA_West_Papua_PB05	500	20	75	106	142	20	79
		HCSA_West_Papua_PB11	900	20	376	302	80	181	700
		HCSA_West_Papua_PB21	1320	20	190	304	160	206	798
		HCSA_West_Papua_PB17	340	60	83	272	328	238	923
		HCSA_West_Papua_PB07	1540	20	225	229	102	138	534

TABLE XXIII
PER-PLOT ROWS, SIMULATIONS 6–10

Sim	Region	Plot	Stems	Trees	Field	Map	Map %	E4	E4 sc.
6	HCSA Riau	HCSA_Riau_RU16	80	20	146	148	102	74	283
		HCSA_Riau_RU06	720	0	170	120	71	0	0
		HCSA_Riau_RU04	1120	0	202	159	79	0	0
		HCSA_Riau_RU05	960	0	462	237	51	0	0
		HCSA_Riau_RU22	60	40	12	78	649	23	90
		HCSA_Riau_RU12	500	0	195	259	133	0	0
		HCSA_Riau_RU08	720	80	460	119	26	30	114
		HCSA_Riau_RU11	820	0	104	156	150	0	0
		HCSA_Riau_RU15	580	20	114	463	407	30	117
		HCSA_Riau_RU19	420	0	45	91	203	0	0
7	Labrière Rabi	LAB_RAB12h	–	23	534	588	110	213	816
		LAB_RAB17h	–	14	458	500	109	175	672
		LAB_RAB18h	–	11	452	581	128	128	492
		LAB_RAB05h	–	19	553	469	85	128	491
		LAB_RAB02h	–	19	483	501	104	146	559
		LAB_RAB16h	–	24	437	471	108	151	577
		LAB_RAB19h	–	22	440	423	96	151	581
		LAB_RAB11h	–	15	496	483	97	154	589
		LAB_RAB25h	–	18	506	522	103	157	604
		LAB_RAB08h	–	19	361	504	140	148	566
8	Labrière Mabounie	LAB_MAB10h	–	4	594	200	34	11	40
		LAB_MAB07h	–	12	426	550	129	88	334
		LAB_MAB04h	–	18	791	659	83	209	788
		LAB_MAB03h	–	0	575	187	33	0	0
		LAB_MAB01h	–	23	565	489	87	134	505
		LAB_MAB08h	–	22	501	576	115	150	566
		LAB_MAB05h	–	9	756	717	95	106	400
		LAB_MAB12h	–	16	296	544	184	121	456
		LAB_MAB06h	–	14	533	548	103	129	488
		LAB_MAB02h	–	12	521	624	120	112	422
9	HCSA Jambi	HCSA_Jambi_JB15	540	60	54	106	194	10	39
		HCSA_Jambi_JB02	960	20	180	298	165	145	558
		HCSA_Jambi_JB05	1300	20	180	538	298	232	893
		HCSA_Jambi_JB19	560	0	28	84	304	0	0
		HCSA_Jambi_JB12	1180	0	291	261	90	0	0
		HCSA_Jambi_JB27	340	20	719	189	26	133	511
		HCSA_Jambi_JB14	1420	20	294	116	39	4	16
		HCSA_Jambi_JB18	580	40	222	553	249	132	508
		HCSA_Jambi_JB28	160	0	613	117	19	0	0
		HCSA_Jambi_JB30	380	0	292	409	140	0	0
10	Labrière Mondah	LAB_MON19h	–	3	6	9	145	0	1
		LAB_MON21h	–	3	4	6	146	0	1
		LAB_MON22h	–	9	498	453	91	32	122
		LAB_MON23h	–	22	222	288	130	49	185
		LAB_MON10h	–	14	178	212	119	10	37
		LAB_MON20h	–	14	121	142	117	11	41
		LAB_MON13h	–	17	427	534	125	74	281
		LAB_MON09h	–	0	4	5	133	0	0
		LAB_MON11h	–	23	60	59	98	6	22
		LAB_MON01h	–	20	44	49	112	4	15

Data and Code Availability

All field sources are public: HCSA Indonesia (CC BY-SA 4.0) [60], GEO-TREES [61], Labrière *et al.* (CC0) [62], and the SAFE tree census [89]. Satellite data were read from the Microsoft Planetary Computer [59] and Google Earth Engine. The pipeline (Python; pandas, LightGBM, rasterio, pystac) is in the project repository, including the ledger, the runner, the right-grid builder, and the evaluation scripts. Every reported number is regenerated by the modules `right_grid`, `right_grid_seq`, `right_grid_uncertainty`, `extract_left`, `evaluate`, `evaluate_seq`, `improve`, `improve_seq`, `window_sensitivity`, `cloud_climatology`, `cloud_analysis`, `robust`, `project_uncertainty`, `systems_experiments`, together with `extract_gee`, `accuracy_gee`, `conformal_crediting`, `crediting_frontier`, `crediting_robustness`, `extract_chm`, `tree_count`, `tree_benchmark`, `simulations`, `gain_loss`, `reliability`, and `figures`, of the `estimation` package. Wood density is from the GWDD copy shipped with BIOMASS v2.1.11 [20].

References

[1] J. Chave *et al.*, "Improved allometric models to estimate the above-ground biomass of tropical trees," *Global Change Biol.*, vol. 20, no. 10, pp. 3177–3190, 2014, doi: 10.1111/gcb.12629.

[2] Verra, *VM0047 Afforestation, Reforestation, and Revegetation*, ver. 1.1, May 2025. [Online]. Available: https://verra.org/wp-content/uploads/2025/05/VM0047-v1.1_Final.pdf

[3] R. Dubayah *et al.*, "The Global Ecosystem Dynamics Investigation: High-resolution laser ranging of the Earth's forests and topography," *Sci. Remote Sens.*, vol. 1, Art. no. 100002, 2020.

[4] M. Santoro *et al.*, "The global forest above-ground biomass pool for 2010 estimated from high-resolution satellite observations," *Earth Syst. Sci. Data*, vol. 13, pp. 3927–3950, 2021.

[5] S. A. Spawn, C. C. Sullivan, T. J. Lark, and H. K. Gibbs, "Harmonized global maps of above and belowground biomass carbon density in the year 2010," *Sci. Data*, vol. 7, Art. no. 112, 2020.

[6] P. Ploton *et al.*, "Spatial validation reveals poor predictive performance of large-scale ecological mapping models," *Nat. Commun.*, vol. 11, Art. no. 4540, 2020.

[7] D. R. Roberts *et al.*, "Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure," *Ecography*, vol. 40, pp. 913–929, 2017.

[8] M. Zaffran, A. Dieuleveut, J. Josse, and Y. Romano, "Conformal prediction with missing values," in *Proc. 40th Int. Conf. Machine Learning (ICML)*, PMLR vol. 202, 2023, pp. 40578–40604.

[9] Y. Romano, E. Patterson, and E. Candès, "Conformalized quantile regression," in *Adv. Neural Inf. Process. Syst. 32*, 2019, pp. 3543–3553.

[10] N. Neverova, C. Wolf, G. Taylor, and F. Nebout, "ModDrop: Adaptive multi-modal gesture recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, 2016, doi: 10.1109/TPAMI.2015.2461544.

[11] C.-E. Särndal, B. Swensson, and J. Wretman, *Model Assisted Survey Sampling*. New York: Springer, 1992.

[12] G. Ståhl *et al.*, "Use of models in large-area forest surveys: Comparing model-assisted, model-based and hybrid estimation," *Forest Ecosyst.*, vol. 3, art. 5, 2016, doi: 10.1186/s40663-016-0064-9.

[13] M. Réjou-Méchain *et al.*, "Local spatial structure of forest biomass and its consequences for remote sensing of carbon stocks," *Biogeosciences*, vol. 11, pp. 6827–6840, 2014, doi: 10.5194/bg-11-6827-2014.

[14] A. Araza *et al.*, "A comprehensive framework for assessing the accuracy and uncertainty of global above-ground biomass maps," *Remote Sens. Environ.*, vol. 272, art. 112917, 2022, doi: 10.1016/j.rse.2022.112917.

[15] E. Rutishauser *et al.*, "Generic allometric models including height best estimate forest biomass and carbon stocks in Indonesia," *Forest Ecol. Manag.*, vol. 307, pp. 219–225, 2013, doi: 10.1016/j.foreco.2013.07.013.

[16] K. Mokany, R. J. Raison, and A. S. Prokushkin, "Critical analysis of root:shoot ratios in terrestrial biomes," *Global Change Biol.*, vol. 12, pp. 84–96, 2006, doi: 10.1111/j.1365-2486.2005.001043.x.

[17] S. Haber and W. S. Stornetta, "How to time-stamp a digital document," *J. Cryptology*, vol. 3, no. 2, pp. 99–111, 1991.

[18] M. Blum, "Coin flipping by telephone: A protocol for solving impossible problems," *ACM SIGACT News*, vol. 15, no. 1, pp. 23–27, 1983.

[19] A. Blum and M. Hardt, "The Ladder: A reliable leaderboard for machine learning competitions," in *Proc. 32nd Int. Conf. Machine Learning (ICML)*, PMLR vol. 37, 2015, pp. 1006–1014.

[20] M. Réjou-Méchain, A. Tanguy, C. Piponiot, J. Chave, and B. Hérault, "BIOMASS: An R package for estimating above-ground biomass and its uncertainty in tropical forests," *Methods Ecol. Evol.*, vol. 8, no. 9, pp. 1163–1167, 2017.

[21] G. Rosoman, S. S. Sheun, C. Opal, P. Anderson, and R. Trapshah, Eds., *The HCS Approach Toolkit*, ver. 2.0, Module 4. Singapore: HCS Approach Steering Group, 2017.

[22] T. A. P. West *et al.*, "Action needed to make carbon offsets from forest conservation work for climate change mitigation," *Science*, vol. 381, no. 6660, pp. 873–877, 2023.

[23] E. T. A. Mitchard *et al.*, "Using satellite radar backscatter to predict above-ground woody biomass: A consistent relationship across four different African landscapes," *Geophys. Res. Lett.*, vol. 36, Art. no. L23401, 2009.

[24] S. Mermoz, M. Réjou-Méchain, L. Villard, T. Le Toan, V. Rossi, and S. Gourlet-Fleury, "Decrease of L-band SAR backscatter with biomass of dense forests," *Remote Sens. Environ.*, vol. 159, pp. 307–317, 2015.

[25] L. Duncanson *et al.*, "Aboveground biomass density models for NASA's Global Ecosystem Dynamics Investigation (GEDI) lidar mission," *Remote Sens. Environ.*, vol. 270, Art. no. 112845, 2022.

[26] Chloris Geospatial, "Global aboveground biomass stock and change, 2003–2019," distributed by the Microsoft Planetary Computer. [Online]. Available: https://planetarycomputer.microsoft.com/dataset/chloris-biomass

[27] G. P. Asner and J. Mascaro, "Mapping tropical forest carbon: Calibrating plot estimates to a simple LiDAR metric," *Remote Sens. Environ.*, vol. 140, pp. 614–624, 2014.

[28] G. Ke *et al.*, "LightGBM: A highly efficient gradient boosting decision tree," in *Adv. Neural Inf. Process. Syst. 30*, 2017, pp. 3146–3154.

[29] N. Duan, "Smearing estimate: A nonparametric retransformation method," *J. Amer. Statist. Assoc.*, vol. 78, no. 383, pp. 605–610, 1983.

[30] V. Vovk, A. Gammerman, and G. Shafer, *Algorithmic Learning in a Random World*. New York, NY, USA: Springer, 2005.

[31] L. I.-K. Lin, "A concordance correlation coefficient to evaluate reproducibility," *Biometrics*, vol. 45, no. 1, pp. 255–268, 1989.

[32] A. M. Wilson and W. Jetz, "Remotely sensed high-resolution global cloud dynamics for predicting ecosystem and biodiversity distributions," *PLoS Biol.*, vol. 14, no. 3, e1002415, 2016, doi: 10.1371/journal.pbio.1002415.

[33] M. Sudmanns, D. Tiede, H. Augustin, and S. Lang, "Assessing global Sentinel-2 coverage dynamics and data availability for operational Earth observation (EO) applications using the EO-Compass," *Int. J. Digit. Earth*, vol. 13, no. 7, pp. 768–784, 2020, doi: 10.1080/17538947.2019.1572799.

[34] J. Li and D. P. Roy, "A global analysis of Sentinel-2A, Sentinel-2B and Landsat-8 data revisit intervals and implications for terrestrial monitoring," *Remote Sens.*, vol. 9, no. 9, art. 902, 2017, doi: 10.3390/rs9090902.

[35] Z. Zhu and C. E. Woodcock, "Object-based cloud and cloud shadow detection in Landsat imagery," *Remote Sens. Environ.*, vol. 118, pp. 83–94, 2012, doi: 10.1016/j.rse.2011.10.028.

[36] S. Skakun *et al.*, "Cloud Mask Intercomparison eXercise (CMIX): An evaluation of cloud masking algorithms for Landsat 8 and Sentinel-2," *Remote Sens. Environ.*, vol. 274, art. 112990, 2022, doi: 10.1016/j.rse.2022.112990.

[37] V. J. Pasquarella, C. F. Brown, W. Czerwinski, and W. J. Rucklidge, "Comprehensive quality assessment of optical satellite imagery using weakly supervised video learning," in *Proc. IEEE/CVF CVPR Workshops*, 2023, pp. 2125–2135, doi: 10.1109/CVPRW59228.2023.00206.

[38] A. I. Flores-Anderson *et al.*, "Spatial and temporal availability of cloud-free optical observations in the tropics to monitor deforestation," *Sci. Data*, vol. 10, art. 550, 2023, doi: 10.1038/s41597-023-02439-x.

[39] J. C. White *et al.*, "Pixel-based image compositing for large-area dense time series applications and science," *Can. J. Remote Sens.*, vol. 40, no. 3, pp. 192–212, 2014, doi: 10.1080/07038992.2014.945827.

[40] M. Claverie *et al.*, "The Harmonized Landsat and Sentinel-2 surface reflectance data set," *Remote Sens. Environ.*, vol. 219, pp. 145–161, 2018, doi: 10.1016/j.rse.2018.09.002.

[41] J. Reiche *et al.*, "Combining satellite data for better tropical forest monitoring," *Nature Clim. Change*, vol. 6, pp. 120–122, 2016, doi: 10.1038/nclimate2919.

[42] A. Meraner, P. Ebel, X. X. Zhu, and M. Schmitt, "Cloud removal in Sentinel-2 imagery using a deep residual neural network and SAR-optical data fusion," *ISPRS J. Photogramm. Remote Sens.*, vol. 166, pp. 333–346, 2020, doi: 10.1016/j.isprsjprs.2020.05.013.

[43] J. Lei, M. G'Sell, A. Rinaldo, R. J. Tibshirani, and L. Wasserman, "Distribution-free predictive inference for regression," *J. Amer. Statist. Assoc.*, vol. 113, no. 523, pp. 1094–1111, 2018, doi: 10.1080/01621459.2017.1307116.

[44] H. Papadopoulos, A. Gammerman, and V. Vovk, "Normalized nonconformity measures for regression conformal prediction," in *Proc. IASTED Int. Conf. Artificial Intelligence and Applications*, 2008, pp. 64–69.

[45] H. Boström and U. Johansson, "Mondrian conformal regressors," in *Proc. COPA*, PMLR vol. 128, 2020, pp. 114–133.

[46] R. J. Tibshirani, R. F. Barber, E. Candès, and A. Ramdas, "Conformal prediction under covariate shift," in *Adv. Neural Inf. Process. Syst. 32*, 2019, pp. 2530–2540.

[47] H. Mao, R. Martin, and B. J. Reich, "Valid model-free spatial prediction," *J. Amer. Statist. Assoc.*, vol. 119, no. 546, pp. 904–914, 2024, doi: 10.1080/01621459.2022.2147531.

[48] M. Kuronen, J. Räty, P. Packalen, and M. Myllymäki, "Uncertainty quantification for forest attribute maps with conformal prediction and k-nearest neighbor method," *Remote Sens. Environ.*, vol. 325, art. 114758, 2025, doi: 10.1016/j.rse.2025.114758.

[49] G. Singh, G. Moncrieff, Z. Venter, K. Cawse-Nicholson, J. Slingsby, and T. B. Robinson, "Uncertainty quantification for probabilistic machine learning in Earth observation using conformal prediction," *Sci. Rep.*, vol. 14, art. 16166, 2024, doi: 10.1038/s41598-024-65954-w.

[50] L. Duncanson, M. Disney, J. Armston, J. Nickeson, D. Minor, and F. Camacho, Eds., *Aboveground Woody Biomass Product Validation Good Practices Protocol*, ver. 1.0, CEOS WGCV Land Product Validation Subgroup, 2021, doi: 10.5067/doc/ceoswgcv/lpv/agb.001.

[51] L. Duncanson *et al.*, "The importance of consistent global forest aboveground biomass product validation," *Surv. Geophys.*, vol. 40, no. 4, pp. 979–999, 2019, doi: 10.1007/s10712-019-09538-8.

[52] W. G. Cochran, *Sampling Techniques*, 3rd ed. New York: Wiley, 1977.

[53] E. W. Mauya *et al.*, "Effects of field plot size on prediction accuracy of aboveground biomass in airborne laser scanning-assisted inventories in tropical rain forests of Tanzania," *Carbon Balance Manag.*, vol. 10, art. 10, 2015, doi: 10.1186/s13021-015-0021-x.

[54] J. Breidenbach, R. E. McRoberts, and R. Astrup, "Empirical coverage of model-based variance estimators for remote sensing assisted estimation of stand-level timber volume," *Remote Sens. Environ.*, vol. 173, pp. 274–281, 2016, doi: 10.1016/j.rse.2015.07.026.

[55] A. M. J.-C. Wadoux and G. B. M. Heuvelink, "Scientists yet to consider spatial correlation in assessing uncertainty of spatial averages and totals," *Int. J. Appl. Earth Obs. Geoinf.*, vol. 139, art. 104472, 2025, doi: 10.1016/j.jag.2025.104472.

[56] J. Dean and S. Ghemawat, "MapReduce: Simplified data processing on large clusters," *Commun. ACM*, vol. 51, no. 1, pp. 107–113, 2008.

[57] D. Karger, E. Lehman, T. Leighton, R. Panigrahy, M. Levine, and D. Lewin, "Consistent hashing and random trees," in *Proc. 29th ACM Symp. Theory Comput.*, 1997, pp. 654–663.

[58] R. L. Graham, "Bounds on multiprocessing timing anomalies," *SIAM J. Appl. Math.*, vol. 17, no. 2, pp. 416–429, 1969.

[59] Microsoft Open Source, M. McFarland, R. Emanuele, D. Morris, and T. Augspurger, "microsoft/PlanetaryComputer: October 2022," Zenodo, 2022, doi: 10.5281/zenodo.7261897.

[60] High Carbon Stock Approach, "Forest field plot data for indicative HCS mapping, Indonesia," Fair Forward/GIZ and JKPP, 2023, CC BY-SA 4.0. [Online]. Available: https://huggingface.co/datasets/fair-forward/hcsa_indonesia

[61] GEO-TREES, "Biomass reference plot data," IIASA, accessed Oct. 2026. [Online]. Available: https://geo-trees.nodes.iiasa.ac.at/page/data

[62] N. Labrière *et al.*, "In situ reference datasets from the TropiSAR and AfriSAR campaigns in support of upcoming spaceborne biomass missions," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 10, pp. 3617–3627, 2018, doi: 10.1109/JSTARS.2018.2851606.

[63] Badan Standardisasi Nasional, *SNI 7724:2011 Pengukuran dan penghitungan cadangan karbon – Pengukuran lapangan untuk penaksiran cadangan karbon hutan (ground based forest carbon accounting)*. Jakarta: BSN, 2011.

[64] Ministry of Environment and Forestry, Republic of Indonesia, *National Forest Reference Level for Deforestation, Forest Degradation and Enhancement of Forest Carbon Stock*, submission to the UNFCCC, 2022. [Online]. Available: https://redd.unfccc.int/files/2nd_frl_indonesia_final_submit.pdf

[65] IPCC, *2006 IPCC Guidelines for National Greenhouse Gas Inventories, Vol. 4: Agriculture, Forestry and Other Land Use*, Ch. 2 and 4, H. S. Eggleston *et al.*, Eds. Hayama, Japan: IGES, 2006.

[66] IPCC, *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Vol. 4*, Ch. 4 Forest Land, Table 4.4 (Updated). Geneva: IPCC, 2019.

[67] IPCC, *2006 IPCC Guidelines for National Greenhouse Gas Inventories, Vol. 1*, Ch. 3 Uncertainties. Hayama, Japan: IGES, 2006.

[68] UNFCCC CDM Executive Board, *A/R Methodological Tool 14: Estimation of Carbon Stocks and Change in Carbon Stocks of Trees and Shrubs in A/R CDM Project Activities*, ver. 04.2.

[69] UNFCCC CDM Executive Board, *A/R Methodological Tool 03: Calculation of the Number of Sample Plots for Measurements within A/R CDM Project Activities*, ver. 2.1.0, EB 58, Annex 15.

[70] Gold Standard, *Afforestation/Reforestation GHG Emissions Reduction & Sequestration Methodology*, ver. 2.1, May 2024.

[71] A. E. Zanne *et al.*, "Data from: Towards a worldwide wood economics spectrum," Dryad, 2009, doi: 10.5061/dryad.234.

[72] J. Chave, D. Coomes, S. Jansen, S. L. Lewis, N. G. Swenson, and A. E. Zanne, "Towards a worldwide wood economics spectrum," *Ecol. Lett.*, vol. 12, no. 4, pp. 351–366, 2009, doi: 10.1111/j.1461-0248.2009.01285.x.

[73] T. M. Basuki, P. E. van Laake, A. K. Skidmore, and Y. A. Hussin, "Allometric equations for estimating the above-ground biomass in tropical lowland Dipterocarp forests," *Forest Ecol. Manag.*, vol. 257, pp. 1684–1694, 2009, doi: 10.1016/j.foreco.2009.01.027.

[74] R. C. Goodman *et al.*, "Amazon palm biomass and allometry," *Forest Ecol. Manag.*, vol. 310, pp. 994–1004, 2013, doi: 10.1016/j.foreco.2013.09.045.

[75] K. Lewis *et al.*, "An assessment of oil palm plantation aboveground biomass stocks on tropical peat using destructive and non-destructive methods," *Sci. Rep.*, vol. 10, art. 2230, 2020, doi: 10.1038/s41598-020-58982-9.

[76] J. Chave, R. Condit, S. Aguilar, A. Hernandez, S. Lao, and R. Perez, "Error propagation and scaling for tropical forest biomass estimates," *Phil. Trans. R. Soc. B*, vol. 359, pp. 409–420, 2004, doi: 10.1098/rstb.2003.1425.

[77] S. C. Thomas and A. R. Martin, "Carbon content of tree tissues: A synthesis," *Forests*, vol. 3, no. 2, pp. 332–352, 2012, doi: 10.3390/f3020332.

[78] Verra, *VMD0017 Estimation of Uncertainty for REDD+ Project Activities*, ver. 2.2, Sep. 2020.

[79] J. Tolan *et al.*, "Very high resolution canopy height maps from RGB imagery using self-supervised vision transformer and convolutional decoder trained on aerial lidar," *Remote Sens. Environ.*, vol. 300, art. 113888, 2024, doi: 10.1016/j.rse.2023.113888.

[80] N. Lang, W. Jetz, K. Schindler, and J. D. Wegner, "A high-resolution canopy height model of the Earth," *Nature Ecol. Evol.*, vol. 7, pp. 1778–1789, 2023, doi: 10.1038/s41559-023-02206-6.

[81] R. Dubayah *et al.*, "GEDI launches a new era of biomass inference from space," *Environ. Res. Lett.*, vol. 17, art. 095001, 2022, doi: 10.1088/1748-9326/ac8694.

[82] M. C. Hansen *et al.*, "High-resolution global maps of 21st-century forest cover change," *Science*, vol. 342, pp. 850–853, 2013, doi: 10.1126/science.1244693.

[83] C. J. Tucker, "Red and photographic infrared linear combinations for monitoring vegetation," *Remote Sens. Environ.*, vol. 8, no. 2, pp. 127–150, 1979, doi: 10.1016/0034-4257(79)90013-0.

[84] S. C. Popescu and R. H. Wynne, "Seeing the trees in the forest: Using lidar and multispectral data fusion with local filtering and variable window size for estimating tree height," *Photogramm. Eng. Remote Sens.*, vol. 70, no. 5, pp. 589–604, 2004.
[85] T. Jucker *et al.*, "Allometric equations for integrating remote sensing imagery into forest monitoring programmes," *Global Change Biol.*, vol. 23, no. 1, pp. 177–190, 2017, doi: 10.1111/gcb.13388.
[86] G. Grassi *et al.*, "Applying the conservativeness principle to REDD to deal with the uncertainties of the estimates," *Environ. Res. Lett.*, vol. 3, art. 035005, 2008, doi: 10.1088/1748-9326/3/3/035005.
[87] T. Neeff, "What is the risk of overestimating emission reductions from forests – and what can be done about it?" *Climatic Change*, vol. 166, art. 26, 2021, doi: 10.1007/s10584-021-03079-z.
[88] R. Dunn, L. Wasserman, and A. Ramdas, "Distribution-free prediction sets for two-layer hierarchical models," *J. Amer. Statist. Assoc.*, vol. 118, no. 544, pp. 2491–2502, 2023, doi: 10.1080/01621459.2022.2060112.
[89] M. Svátek *et al.*, "Tree census data from the SAFE Project 2011–2020," Zenodo, 2025, doi: 10.5281/zenodo.14882506.
[90] R. M. Ewers *et al.*, "A large-scale forest fragmentation experiment: The Stability of Altered Forest Ecosystems Project," *Philos. Trans. R. Soc. B*, vol. 366, pp. 3292–3302, 2011, doi: 10.1098/rstb.2011.0049.
[91] H. Meyer and E. Pebesma, "Predicting into unknown space? Estimating the area of applicability of spatial prediction models," *Methods Ecol. Evol.*, vol. 12, pp. 1620–1633, 2021, doi: 10.1111/2041-210X.13650.
[92] I. Shishlov and V. Bellassen, "Review of the experience with monitoring uncertainty requirements in the Clean Development Mechanism," *Climate Policy*, vol. 16, pp. 703–731, 2016, doi: 10.1080/14693062.2015.1046414.
[93] G. Badgley *et al.*, "Systematic over-crediting in California's forest carbon offsets program," *Global Change Biol.*, vol. 28, pp. 1433–1445, 2022, doi: 10.1111/gcb.15943.
[94] D. Requena Suarez *et al.*, "Estimating aboveground net biomass change for tropical and subtropical forests: Refinement of IPCC default rates using forest plot data," *Global Change Biol.*, vol. 25, pp. 3609–3624, 2019, doi: 10.1111/gcb.14767.
[95] L.-C. Zhang and T. Tuoto, "Conformal confidence intervals with an application to small area estimation," arXiv:2608.02766, 2026.
[96] S. Kaul and G. J. Gordon, "Meta-analysis with untrusted data," arXiv:2407.09387, 2024.
[97] A. Berninger, S. Lohberger, M. Stängel, and F. Siegert, "SAR-based estimation of above-ground biomass and its changes in tropical forests of Kalimantan using L- and C-band," *Remote Sens.*, vol. 10, no. 6, art. 831, 2018, doi: 10.3390/rs10060831.
[98] T. Jucker *et al.*, "Estimating aboveground carbon density and its uncertainty in Borneo's structurally complex tropical forests using airborne laser scanning," *Biogeosciences*, vol. 15, pp. 3811–3830, 2018, doi: 10.5194/bg-15-3811-2018.
[99] E. Hyyppä *et al.*, "Under-canopy UAV laser scanning for accurate forest field measurements," *ISPRS J. Photogramm. Remote Sens.*, vol. 164, pp. 41–60, 2020, doi: 10.1016/j.isprsjprs.2020.03.021.
[100] X. Liang, H. Yao, H. Qi, and X. Wang, "Forest in situ observations through a fully automated under-canopy unmanned aerial vehicle," *Geo-spat. Inf. Sci.*, vol. 27, pp. 983–999, 2024, doi: 10.1080/10095020.2024.2322765.
[101] X. Zhou *et al.*, "Swarm of micro flying robots in the wild," *Sci. Robot.*, vol. 7, art. eabm5954, 2022, doi: 10.1126/scirobotics.abm5954.
[102] A. Ferraz *et al.*, "Carbon storage potential in degraded forests of Kalimantan, Indonesia," *Environ. Res. Lett.*, vol. 13, art. 095001, 2018, doi: 10.1088/1748-9326/aad782.
[103] J. Chave *et al.*, "Regional and phylogenetic variation of wood density across 2456 Neotropical tree species," *Ecol. Appl.*, vol. 16, pp. 2356–2367, 2006, doi: 10.1890/1051-0761(2006)016[2356:RAPVOW]2.0.CO;2.
[104] J. Williams *et al.*, "Monitoring early-successional trees for tropical forest restoration using low-cost UAV-based species classification," *Front. For. Glob. Change*, vol. 5, art. 876448, 2022, doi: 10.3389/ffgc.2022.876448.
[105] G. P. Asner *et al.*, "High-resolution forest carbon stocks and emissions in the Amazon," *Proc. Natl. Acad. Sci. USA*, vol. 107, pp. 16738–16742, 2010, doi: 10.1073/pnas.1004875107.
[106] D. Wang *et al.*, "Estimating aboveground biomass of the mangrove forests on northeast Hainan Island in China using an upscaling method from field plots, UAV-LiDAR data and Sentinel-2 imagery," *Int. J. Appl. Earth Obs. Geoinf.*, vol. 85, art. 101986, 2020, doi: 10.1016/j.jag.2019.101986.
[107] IPCC, *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Vol. 4*, Ch. 2, Sec. 2.3.1. Geneva: IPCC, 2019.